Finite-Sample Valid Tests for Bounded Treatment Effects

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Abstract

Randomization tests deliver exact finite-sample inference but are typically limited to sharp null hypotheses that identify every unit's missing potential outcome. This paper develops finite-sample valid randomization tests for a broader class of null hypotheses that impose bounds on individual treatment effects. Tests of these hypotheses have broad applications including testing instrumental variable monotonicity, testing whether proposed interventions harm any segment of a population, and assessing whether distributions of outcomes are sufficiently similar across experimental conditions to justify generalization. The key insight is identifying a family of "effect-monotone" statistics for which the least favorable distribution under a bounded null places each unit's treatment effect at the boundary of the constraint, so computing the randomization distribution at this boundary yields conservative p-values. By embedding the two-sample problem in the potential outcomes framework, these results also yield new finite-sample valid tests for bounds on quantile differences between two distributions, including stochastic dominance and distributional closeness.

1 Introduction

Randomization testing has a long history in statistics, tracing back to Fisher (1935) and his celebrated "lady tasting tea" experiment. The core ideas were refined and extended by subsequent researchers (Pitman, 1937; Welch, 1937; Wilcoxon, 1945; Lehmann, 1949; Kempthorne, 1952), and today randomization inference occupies a valuable niche in empirical econometrics. The appeal is straightforward: these procedures deliver finite-sample control of Type I error without relying on parametric assumptions or large-sample approximations. In settings where asymptotic theory may falter, because samples are small, clusters are few, or sources of identifying variation are limited, the exactness of randomization inference offers what Ritzwoller et al. (2025) aptly describe as "comforting, if not liberating" guarantees. However, it is widely recognized that randomization-style methods are not universally applicable. Classical randomization methods require the ability to simulate the distribution of a test statistic under the null hypothesis, and that ability is typically confined to a narrow class of null hypotheses.

Before discussing this limitation, it is helpful to clarify what I mean by a randomization test. The terminology in this area is notoriously unsettled: "randomization inference," "permutation tests," "rerandomization tests," "conditional randomization tests," and "group-invariance tests"

have all been used to describe closely related, but not identical, procedures. Valuable efforts to impose a taxonomy on this family of methods already exist (Zhang and Zhao, 2023), so I will not add to that enterprise here. Instead, I offer a broad working definition that captures the procedures I aim to extend.

What unites this family of tests is a shared logic: validity rests on the ability to simulate, or otherwise characterize, a known distribution of a test statistic under the null hypothesis, possibly conditional on an information set coarser than the full observed sample. Any parametric null trivially satisfies this requirement, since the null pins down the sampling distribution. The more interesting cases arise when the null is nonparametric and the reference distribution derives from known features of the data-generating process itself, such as a randomization or assignment mechanism whose law is specified by the experimental design.

This limitation on the settings in which randomization logic applies is often articulated in one of two closely related ways. The first is a *group invariance* or *randomization hypothesis* condition: under the null, the distribution of the observed data is invariant to a known group of transformations (typically permutations), which implies that the conditional distribution of the data over this group of transformations is known and can be used to form exact p-values (Romano, 1990; Lehmann and Romano, 2022; Ritzwoller et al., 2025). The second is the "sharp null" condition in the potential outcomes framework: the null must be sufficiently informative that, given the observed data, one can impute at least a non-empty set of the unobserved potential outcomes (almost surely) and thereby compute the randomization distribution of the statistic under the null. I follow Zhang and Zhao (2023) and call a null *sharp* if, in a potential outcomes model, the null implies imputability of a non-empty subset of unobserved potential outcomes from observed outcomes (almost surely). In the canonical example of a completely randomized experiment, the sharp null of no individual treatment effects, $Y_i(1) = Y_i(0)$ for all i, renders the entire schedule of potential outcomes imputable, so the reference distribution of any statistic can be computed exactly by re-randomizing the assignment.

Many hypotheses of central economic interest are not sharp. Moment restrictions such as zero average treatment effect, distributional restrictions such as equality of medians, and many forms of partial identification and inequality restrictions generally do not pin down missing potential outcomes at the unit level. As a result, extending randomization inference beyond sharp nulls has been an active theme in the literature. A common strategy is to retain an exact randomization test for some sharp null while establishing that the same procedure is *asymptotically* valid for a weaker, non-sharp null. This perspective traces back to early work on permutation and randomization tests for weak null hypotheses (Romano, 1990; Janssen, 1997) and remains central in more recent contributions (Chung and Romano, 2013; DiCiccio and Romano, 2017; Canay et al., 2017; Wu and Ding, 2021; Chung and Olivares, 2025; Ritzwoller et al., 2025). These approaches can be powerful and broadly applicable, but they typically trade away the finite-sample exactness that motivates randomization inference in the first place. When asymptotics are unreliable, this trade-off can be consequential.

This paper contributes to a broader project aimed at adapting the underlying logic of random-

ization tests to obtain finite-sample valid procedures for non-sharp null hypotheses. The focus here is a class of nulls that I call *bounded treatment effect nulls*. In a binary treatment potential outcomes framework, these nulls restrict each unit's treatment effect $\tau_i := Y_i(1) - Y_i(0)$ to lie within known bounds that may depend on observable characteristics and (at most) one potential outcome. For known upper (u) and lower (ℓ) functions chosen by the researcher, the bounded-effect null hypotheses take the one-sided forms

$$H_0^\ell: \quad P(\tau_i \geq \ell(Y_i(0), Z_i) \text{ for all } i) = 1, \qquad H_0^u: \quad P(\tau_i \leq u(Y_i(0), Z_i) \text{ for all } i) = 1,$$

and the two-sided version

$$H_0^{\ell,u}: \quad P(\ell(Y_i(0), Z_i) \leq \tau_i \leq u(Y_i(0), Z_i) \text{ for all } i) = 1.$$

These hypotheses are generally non-sharp: for any realized observation, there are typically infinitely many counterfactual potential outcomes consistent with the bounds. Nevertheless, the bounds impose *pointwise* restrictions on the potential outcome schedule, and this structure can be exploited to recover finite-sample validity.

Bounded-effect hypotheses arise naturally when economic questions are framed in terms of harm constraints, minimum benefit requirements, or practical relevance thresholds rather than mean effects alone. Three examples illustrate the range.

First, a "harmless treatment" hypothesis sets $\ell(y, z) \equiv 0$ and asks whether the intervention is weakly non-harmful for every unit in the sample. This type of requirement is common in policy contexts where a Pareto-style welfare concern is relevant, and it can also be useful as a specification check when non-negativity of effects is invoked as an identifying assumption (Bai et al., 2025).

Second, bounds that scale with the untreated outcome accommodate settings where treatment effects are naturally expressed in proportional terms. For example, setting $\ell(y, z) = \rho y$ for $\rho \in [0, 1]$ allows testing whether an intervention improves outcomes by at least a fixed percentage, a framing that is particularly relevant for outcomes like savings rates or earnings where absolute magnitudes vary widely across individuals. This flexibility connects to a broader literature emphasizing the policy relevance of treatment effect heterogeneity conditional on the untreated potential outcome (Heckman et al., 1997). Third, one-sided bounds naturally encode monotonicity restrictions in instrumental variables designs. By reinterpreting the "outcome" as treatment uptake and the "treatment" as a randomized instrument, a lower bound of zero corresponds to the standard monotonicity assumption (e.g. no defiers) (Imbens and Angrist, 1994).

Two-sided bounds also have applications. A particularly transparent application is a test of *relevant differences* in the sense of Wellek (2010), where $\ell(y, z) = -\delta$ and $u(y, z) = \delta$ for an economically meaningful threshold $\delta > 0$. This shifts attention from rejecting an exact zero-effect hypothesis to detecting effects that are substantively large enough to matter, while also gaining finite-sample validity.

The core technical idea of the testing procedure is simple to state. A randomization test is valid

if the rejection probability is controlled under the null distribution of the statistic, conditional on an information set that is observable from the sample. When the null is non-sharp, the conditional distribution of the statistic is typically not known because the missing potential outcomes are not identified. The key observation used in this paper is that for bounded-effect nulls, and for a broad class of statistics, one can construct a *least favorable* sharp null hypothesis inside the bounded null such that the induced conditional distribution of the statistic is stochastically worst-case for size control. Computing the reference distribution under this boundary sharp null then yields a conservative (and finite-sample valid) test for the original bounded null.

To make this work, the statistic must interact with unit-level effects in a monotone way. Following Rosenbaum (2002) and Caughey et al. (2023), I focus on *effect-increasing* statistics, for which raising treated outcomes and/or lowering control outcomes cannot decrease the statistic. This class includes many common choices in applied work, including difference-in-means, rank-based statistics, and one-sided Kolmogorov–Smirnov statistics. For one-sided bounded nulls, effect-increasing statistics allow the researcher to select a single degenerate distribution over the potential outcomes at the boundary of the null that delivers the least favorable randomization distribution. For two-sided bounded nulls, I introduce a novel two-sided procedure based on testing two boundary sharp nulls simultaneously in a way that preserves finite-sample validity without relying on Bonferroni-style corrections that can be power limiting.

A second contribution of the paper is to connect bounded treatment effect nulls to a classical setting where randomization tests are routinely used: the two-sample problem. In a standard two-sample design, the analyst observes independent samples from two distributions and is interested in testing the relationship between the two distributions, commonly the equivalence of distributions or equality of means. Many hypotheses of interest in two-sample problems are not sharp, however, including stochastic dominance and approximate distributional equivalence.

I show that the two-sample setting can be embedded in the superpopulation potential outcomes framework through a latent quantile representation. Specifically, for each unit i there exists $Q_i \sim \text{Unif}[0,1]$ such that $Y_i(1) = F^{-1}(Q_i)$ and $Y_i(0) = G^{-1}(Q_i)$, where F and G are the distributions of interest. Under this construction, the unit-level treatment effect $\tau_i = F^{-1}(Q_i) - G^{-1}(Q_i)$ corresponds to the quantile difference at a random quantile, and bounded-effect nulls translate directly into uniform bounds on the quantile difference function. This mapping allows the theorems developed for bounded treatment effects to apply immediately to two-sample hypotheses, including first-order stochastic dominance, dominance with tolerance δ, and two-sided distributional closeness restrictions. To my knowledge, this paper provides the first proof of finite-sample validity for several such randomization-style tests in the two-sample literature.

As previously mentioned, a large literature studies tests that coincide with an exact randomization test for a sharp null but remain valid, typically only asymptotically, for weaker null hypotheses. A smaller but rapidly growing literature seeks finite-sample valid randomization tests beyond sharp nulls. The closest starting point for the present paper is Caughey et al. (2023), who develop finite-sample valid tests for similar bounded treatment effect hypotheses and quantile treatment effect

restrictions in a design-based framework where potential outcomes are treated as fixed. This work has inspired follow-ups that refine computation and extend coverage to additional estimands and settings (Chen and Li, 2024; Su and Li, 2024). In particular Chen and Li (2024) do extend their testing procedure for restrictions on quantiles of the treatment effect distribution to finite population and superpopulation frameworks. While the bounded treatment effect hypotheses are special cases of restrictions on treatment effect quantiles, the testing procedure in this paper allows a broader set of test statistics (and thus, potentially increased power) and novel justification compared to Chen and Li (2024). In the two-sample context, Bugni et al. (2025) have developed a proof of finite-sample randomization-style inference for stochastic dominance using Kolmogorov–Smirnov statistics. Relative to this literature, the present paper makes three main additions: (i) it extends one-sided bounded-effect testing beyond a purely design-based setup to a general potential outcomes framework in which the null is a restriction on a family of probability measures; (ii) it introduces a novel two-sided bounded-effect test that achieves finite-sample validity without reducing to a conservative combination of one-sided tests; and (iii) exploiting the fact that the two-sample problem is a special case of the general binary treatment potential outcomes framework, it provides constructions for a broad set of bounded quantile difference tests in the two-sample problem, establishing finite-sample validity for a likewise broad class of statistics.

Finally, this paper is motivated by the many settings in empirical economics where asymptotic approximations can be misleading because the effective number of independent sources of variation is small. Prominent examples include clustered designs with few clusters (Cameron et al., 2008), regression discontinuity designs where there can be few observations near the cut-off (Canay and Kamat, 2018), shift-share designs where identifying variation comes from a limited number of shocks (Adão et al., 2019), and panel settings with dependence structures that undermine standard large-n heuristics (Spector et al., 2025). Randomization inference has been viewed as a principled alternative in such environments when it applies. However, typically researchers assume mean equivalence of error distributions, or similar, when performing inference on parameters of interest, and are not interested in schemes where sharp nulls are satisfied. Extending finite-sample valid procedures to non-sharp nulls is therefore a prerequisite for bringing the benefits of randomization logic to many of the hypotheses economists most often test.

The remainder of the paper is organized as follows. Section 2 outlines several key motivating examples. Section 3 introduces the general potential outcomes framework and how this includes as special cases several canonical settings of interest. Section 4 provides a sufficient condition for finite-sample valid randomization-style tests based on known upper bounds on conditional tail probabilities, and relates this framing to the traditional group invariance perspective. Section 5 develops finite-sample valid tests for one-sided bounded treatment effect nulls and establishes their implications in the two-sample problem. Section 6 introduces the two-sided bounded treatment effect tests and their two-sample analogues. In Section 7 I conclude.

2 Motivating Examples

Because testing procedures for bounded treatment effect hypotheses have not been widely known, even in large-sample settings, it can be difficult to envision their practical applications. This section therefore presents several motivating examples in which such tests would be useful to empirical economists.

2.1 Test of Monotonicity (and Randomization) in an IV Design

Devoto et al. (2012) study the impact of private household water connections in urban Morocco and their impact on household outcomes using a randomized encouragement design. Households without a private connection were eligible for a program that simplified the connection procedure and offered credit to finance the up-front cost, with program staff facilitating the administrative steps. This randomly assigned encouragement was then treated as an instrument for private water connection to study the causal relationship with outcomes of interest like health, well-being, social integration and time use. Like any instrumental variable design in the local average treatment effect framework, this analysis relies on the assumption of *instrument monotonicity*: the encouragement should not reduce any household's propensity to connect (Imbens and Angrist, 1994). Monotonicity is not innocuous in this setting, however, because some households rely on informal (illegal) connections to neighbors' taps, and being approached about a formal connection may raise enforcement concerns or trigger distrust and avoidance responses. As the authors are particularly interested in the subgroup of households relying on informal connections at baseline, it is important to assess the plausibility of monotonicity in general and in this group in particular, otherwise the standard IV estimand loses its usual interpretation as a complier average causal effect, and the resulting estimates can be difficult to map into policy-relevant causal objects.

Testing monotonicity is a natural application of the finite-sample bounded hypothesis testing procedure developed in this paper, particularly for subgroups of interest in instrumental variable RCT settings where sample sizes are limited by administrative costs. In this study, the analysis sample for households relying on informal connections at baseline is relatively small (fewer than 170 households). Because treatment is assigned at the block level, within-block dependence and limited effective sample sizes can make large-sample approximations unreliable. These features motivate finite-sample procedures that (i) exploit the known randomization distribution induced by the experimental instrument and (ii) remain valid under arbitrary within-block dependence.

To test monotonicity, we can recast the IV design in a potential outcomes framework where the "treatment" is receiving the randomized encouragement and the "outcome" is formal water connection status. Essentially, we are taking the instrument to be "treatment" and the treatment to be the "outcome", for the terminology that is used in the rest of this paper. The individual-level treatment effect τ_i then corresponds to the change in connection status induced by the encouragement instrument. The monotonicity assumption is equivalent to a one-sided bounded treatment effect null of the form $\tau_i \geq 0$ for all i. As we will see, this is testable using the finite-sample valid

randomization test for one-sided bounded treatment effect nulls developed in Section 5. Consistent with much of the literature on testing LATE assumptions (i.e. Kitagawa (2015); Huber and Mellace (2015); Mourifié and Wan (2017); Sun (2023)), because this testing framework requires that treatment (in this case, instrument) assignment be known and independent of potential outcomes, the test is essentially a joint test of monotonicity and randomization. In RCT settings like Devoto et al. (2012), however, randomization is by design, so the test can be interpreted primarily as a test of monotonicity.

2.2 Inference on Lower Bound of Treatment Effects for Microcredit Services

Microcredit services targeting women in developing countries have long been regarded as a promising policy intervention to improve women's social outcomes and reduce poverty in developing communities. Over the past decade, however, a vigorous debate has emerged around potential harms these services may inflict on borrowers. Critics contend that microcredit providers burden households with repayment pressures that crowd out expenditure on essential goods. A series of randomized controlled trials have addressed this concern, but have been mostly concerned with estimating average treatment effects on proxies for harm (Banerjee et al., 2015). Because microcredit is widely believed to generate large positive outcomes in the upper tail of the distribution, average effects may obscure important heterogeneity when assessing potential harms. What may be more important to researchers and policy-makers is whether any segment of the population experiences adverse outcomes, and how severe the worst cases might be. The bounded treatment effect hypothesis tests developed in this paper, and the inference methods they lead to, are a natural tool for such assessments.

I apply these methods to the group-lending microcredit experiment evaluated by Banerjee et al. (2015), which randomized the phased introduction of a lender across 104 low-income neighborhoods in Hyderabad, India. This setting is well suited for finite-sample tests of "no harm" hypotheses. First, the cluster-randomized design at the neighborhood level combined with heavy-tailed outcomes (profits, borrowing, expenditures) motivates inference procedures that remain valid under the known randomization distribution rather than relying on large-sample approximations. Second, concerns about harm are inherently multivariate: a policy-maker may take comfort in null average effects on aggregate consumption yet still worry about adverse effects on repayment stress, essential consumption categories, or children's time use. Inference on the lower bound of treatmente effects provide a direct way to ask: *how much individual-level harm can we rule out, with high confidence, in this experiment?* Finite-smaple valid procedures are particularly attractive when evalutating outcomes for certain subgroups like households on the extensive margin of entrepreneurship, those starting new businesses or lacking a pre-existing enterprise, where downside risk may concentrate among financially marginal households.

2.3 "No relevant differences" in time-preference elicitation

In experimental and behavioral economics, researchers often examine whether behavior is consistent across different settings. A key question is whether findings from one population generalize to another, or whether cost-saving experimental procedures elicit the same behavior as more expensive alternatives. Brañas-Garza et al. (2023) study whether time preferences elicited from a standard multiple-price-list (MPL) task differ when subjects face (i) real payments, (ii) a one-out-of-ten probabilistic payment scheme (BRIS), or (iii) purely hypothetical payments. The authors seek to determine whether the distributions of elicited preferences are similar across these incentive designs. They investigate this using difference tests for mean treatment effects, equivalence tests on the mean, and tests of distributional equivalence. For assessing whether experimental results under one incentive scheme transfer to another, exact equality of means is not necessarily required (since other moments of the distribution may be informative for particular research questions), nor is exact distributional equivalence. Arguably, researchers are interested in whether elicited preferences are "close" up to a given distance metric between distributions. One useful operationalization is testing whether distributions are close in the sense that quantile differences vary by no more than a research-relevant threshold. This is exactly the test described in Section 6.1.

Finite-sample validity is particularly attractive here as the smallest study conducted by the authors (a Spanish lab experiment with $n = 119$ divided across four treatment arms) is genuinely small, and the outcomes are discrete and bounded, so normal or large-n approximations for equivalence-style testing can be unreliable.

3 Setting

We consider a potential outcomes setting in which a sample of individuals of size n, indexed by two potential outcomes, $Y(0)$ and $Y(1)$ and observable characteristics Z are drawn from a distribution with arbitrary dependence between individuals in the sample. We focus on a binary treatment setting where the random vector of assignment $D \in \{0,1\}^n$ is assigned according to a treatment assignment. Particularly the main results, Theorem 2 and Theorem 3, require that the treatment assignment mechanism is known and conditionally independent of the potential outcomes given the covariates Z. This setting encompasses a wide variety of experimental and observational designs, including completely randomized experiments, stratified randomized experiments, and certain observational studies with known treatment assignment probabilities.

Formally, we define a probability space $(\Omega, \mathcal{F}, P)$ on which we define the random vectors, each of size n, W, D and X.

W is a vector of pre-treatment individual characteristics, aggregating the tuples:

$$W_i = (Y_i(0), Y_i(1), Z_i) \quad \text{for } i = 1, \ldots, n,$$

where $Y_i(0) \in \mathcal{Y}_0$ and $Y_i(1) \in \mathcal{Y}_1$ are the potential outcomes for unit i under control and treatment,

respectively, and $Z_i \in \mathcal{Z}$ is a vector of observed covariates for unit i. The sample space for W is given by $\mathcal{W} = \mathcal{Y}_0^n \times \mathcal{Y}_1^n \times \mathcal{Z}^n$.

D is the vector of treatment assignments, $D = (D_1, \ldots, D_n) \in \{0,1\}^n$, where $D_i = 1$ indicates that unit i is assigned to treatment and $D_i = 0$ indicates assignment to control. The known treatment assignment mechanism is denoted by

$$P(D = d \mid Z) \quad \text{for } d \in \{0,1\}^n.$$

X is the vector of observed data, aggregating the tuples:

$$X_i = (Y_i, D_i, Z_i) \quad \text{for } i = 1, \ldots, n,$$

where $Y_i = Y_i(1)D_i + Y_i(0)(1 - D_i)$ is the observed outcome for unit i. The sample space for X is given by

$$\mathcal{X} = \mathcal{Y}^n \times \{0,1\}^n \times \mathcal{Z}^n.$$

Occasionally, it will be useful to consider the canonical space induced by the observed data, which is given by $(\mathcal{X}, \mathcal{F}^X, P^X)$ where $\mathcal{F}^X = \mathcal{B}_{\mathcal{Y}}^{\otimes n} \otimes 2^{\{0,1\}^n} \otimes \mathcal{B}_{\mathcal{Z}}^{\otimes n}$ is the Borel sigma-algebra on $\mathcal{X}$, and P^X is the pushforward measure of P under the mapping $X : \Omega \to \mathcal{X}$.

The hypotheses that I consider in this paper target the individual level treatment effect, defined by

$$\tau_i = Y_i(1) - Y_i(0).$$

Our statistical model is defined by a set of probability measures $\mathcal{P}$ on $(\Omega, \mathcal{F})$, each corresponding to a different data-generating process. These measures encode assumptions about the joint distribution of potential outcomes, treatment assignments, and covariates.

Any given hypothesis corresponds to a subset of $\mathcal{P}$. For example, the sharp null hypothesis of no treatment effect, $\mathcal{H}_{\text{sharp}} : \{P \in \mathcal{P} : P(Y_i(0) = Y_i(1) \text{ for all } i) = 1\}$, corresponds to the subset of measures where $Y_i(1) = Y_i(0)$ almost surely for all i.

We will consider real-valued test statistics of the obervable data, denoted $T : \mathcal{X} \to \mathbb{R}$. Typically we will write this test statistic as a function of the random vector X or a realization $x \in \mathcal{X}$, i.e. $T(X)$ or $T(x)$, where $x = \{y_i, d_i, z_i\}_{i=1}^n$. Occasionally, however, it will be useful to talk about the components of the vector x and we will explicitly expand the vectors of realized outcomes, treatment assignments, and covariates as $x = (y, d, z)$ where $y = (y_1, \ldots, y_n)$, $d = (d_1, \ldots, d_n)$, and $z = (z_1, \ldots, z_n)$, writing $T(y, d, z) = T(x)$.

3.1 Superpopulation Framework

Note that the setting defined above allows arbitrary dependence between units in the sample, and does not require that the data be drawn from a superpopulation. This is done so that the results in this paper can be applied widely to a large variety of potential outcomes frameworks. However, in many applications it is common to consider a superpopulation framework in which individuals

in the sample, W_i are drawn i.i.d. from a larger population.

To adapt our setting to the superpopulation framework, we restrict $\mathcal{P}$ to distributions under which the sample units are independent and identically distributed. Specifically, let Q be a probability measure on the individual-level space $(\mathcal{W}_1, \mathcal{B}_{\mathcal{W}_1})$, where $\mathcal{W}_1 = \mathcal{Y}_0 \times \mathcal{Y}_1 \times \mathcal{Z}$ is the sample space for a single unit's characteristics $W_i = (Y_i(0), Y_i(1), Z_i)$. The superpopulation framework restricts attention to the subset of probability measures $\mathcal{P}^{\text{sp}} \subseteq \mathcal{P}$ such that the law of $W = (W_1, \ldots, W_n)$ is the n-fold product measure $Q^{\otimes n}$; that is, $W_1, \ldots, W_n \overset{\text{iid}}{\sim} Q$.

The bounded treatment effect null hypotheses will essentially place a restriction on the joint distribution of $(Y_i(0), Y_i(1))$ for each unit i in the sample, and thus will also place restrictions on the superpopulation distribution Q. For example, a one-sided bounded treatment effect null hypothesis of the form $\tau_i \leq \delta$ for all i will restrict Q to the subset of distributions satisfying $Q(Y(1) - Y(0) \leq \delta) = 1$. The testing procedure outlined here can then be used to test null hypotheses of this sort.

3.2 Finite Population Framework

Another important special case is the finite population framework, in which individuals are drawn without replacement from a finite population of size N. Let the finite population consist of units indexed by $j \in \{1, \ldots, N\}$ with characteristics $W_j^{(N)} = (Y_j^{(N)}(0), Y_j^{(N)}(1), Z_j^{(N)})$. Sampling is represented by an indicator vector $S = (S_1, \ldots, S_N) \in \{0,1\}^N$ with $\sum_{j=1}^N S_j = n$, where $S_j = 1$ indicates that population unit j is included in the sample.

To embed this in our framework, we restrict $\mathcal{P}$ to the subclass $\mathcal{P}_{\text{FP}}^{(N)}$ of probability measures for which the sample-level variables $W = (W_1, \ldots, W_n)$ are obtained by reindexing the selected population units, and the treatment assignment mechanism satisfies

$$P(D = d \mid \mathbf{W}^{(N)}, S) = P(D = d \mid Z, S) \qquad \forall d \in \{0,1\}^n.$$

The testing procedures developed in this paper apply directly, requiring only that the assignment distribution is known and conditionally independent of potential outcomes given covariates.

3.3 Design-Based Framework

One important special case of the finite population framework is the design-based framework, in which the potential outcomes and covariates of all units in the finite population are considered fixed, and the only source of randomness is the treatment assignment mechanism. In this setting, we restrict $\mathcal{P}$ so that the corresponding law of W is degenerate at some value of the potential outcomes and covariates for all units in the population.

The bounded treatment effect nulls considered in this paper can be interpreted in the design-based framework as placing restrictions on the possible values of the potential outcomes for each unit in the finite population. For example, a one-sided bounded treatment effect null hypothesis of the form $\tau_i \leq \delta$ for all i would restrict the potential outcomes such that $Y_i(1) - Y_i(0) \leq \delta$ for each

unit i in the finite population.

This special case is the case in which Caughey et al. (2023) develop finite-sample valid tests for bounded treatment effects, and the results in this paper can be seen as a generalization of their work to more general potential outcome and two sample settings including superpopulation frameworks. The one-sided bounded treatment effect test has machinery that is very similar to the one-sided test developed in Caughey et al. (2023), and intuitively works in the more general potential outcome setting that I show here, because the bounded treatmtent effect hypotheses place restrictions on the schedules of potential outcomes in W that can be realized up to a null set. The null hypothesis considered by Caughey et al. (2023) is a set of potential outcome schedules, where the null hypothesis in this paper is, in general, a set of probability measures that place probaility of 1 on the realization of that same set of potential outcome schedules. It is in this sense, that this paper can be seen as a direct extension of Caughey et al. (2023). Arguably the authors' forecast in their paper that this sort of extension should be possible, stating "In this paper, we conduct RI where all potential outcomes are viewed as fixed constants... This is equivalent to conducting inference conditional on all the potential outcomes, and thus requires no model or distributional assumptions on the potential outcomes."

However, explicitly performing this extension to the general potential outcomes setting detailed here returns dividends: First, it provides a straightforward reference for researchers employing the one-sided test in a broad array of settings, with explicit justification of the validity of the approach. Second, as we see in the following section, this general framework nests the two-sample framework and thus provides novel proofs of the existence of a number of finite-sample valid tests for bounded treatment effects in two-sample settings that have not been previously documented in the literature. Finally, to make the contributions of this paper explicit, the two-sided bounded treatment effect test proposed here is novel even in the design-based framework, and thus represents a new contribution to the literature on finite-sample valid tests for bounded treatment effects.

3.4 Two-Sample Problem

The classical two-sample problem is a fundamental setting for randomization tests, where the analyst observes samples drawn independently from two unknown distributions and wishes to test hypotheses about their relationship, most commonly whether the distributions are identical or whether their location parameters differ.

Consider the standard two-sample setting where we observe independent and identically distributed samples from two populations:

$$X_i \overset{iid}{\sim} F \text{ for } i = 1, \ldots, n_X \quad \text{and} \quad Y_j \overset{iid}{\sim} G \text{ for } j = 1, \ldots, n_Y.$$

We now show that this setting is a special case of the superpopulation framework developed in Section 3, which allows us to apply the testing procedures developed below to obtain finite-sample valid tests for bounded quantile treatment effect hypotheses in two-sample problems.

To embed the two-sample problem in our potential outcomes framework, we pool the observations into a single sample of size $n = n_X + n_Y$ and interpret group membership as treatment assignment. Define $D_i = 1$ if unit i belongs to the F-sample and $D_i = 0$ if it belongs to the G-sample. Letting W_j denote the j-th observation from the G-sample, the observed outcome for unit i is then $Y_i = D_i X_i + (1 - D_i)W_i$, consistent with the standard potential outcomes relation $Y_i = D_i Y_i(1) + (1 - D_i)Y_i(0)$.

To complete the potential outcomes structure, we must specify the joint distribution of $(Y_i(0), Y_i(1))$ for each unit. We restrict the class of distributions $\mathcal{P}$ to those where the potential outcomes are generated through a common latent quantile. Specifically, for each unit i, we assume there exists a uniformly distributed random variable $Q_i \sim \text{Uniform}[0, 1]$ such that

$$Y_i(1) = F^{-1}(Q_i) \quad \text{and} \quad Y_i(0) = G^{-1}(Q_i),$$

where F^{-1} and G^{-1} denote the quantile functions of the two distributions. This construction ensures that marginally $Y_i(1) \sim F$ and $Y_i(0) \sim G$, as required. The treatment assignment satisfies $\sum_{i=1}^{n} D_i = n_X$, and we assume it is independent of potential outcomes conditional on covariates (or unconditionally when Z is trivial), consistent with our framework.

Then any hypothesis about the relationship between F and G can be translated into a hypothesis about the joint distribution of $(Y_i(0), Y_i(1))$ across units in our potential outcomes framework.

Under this latent quantile construction, the individual treatment effect for unit i becomes the quantile treatment effect at the random quantile Q_i:

$$\tau_i = Y_i(1) - Y_i(0) = F^{-1}(Q_i) - G^{-1}(Q_i).$$

This representation connects individual-level treatment effects in our framework to the quantile difference function $\Delta(q) := F^{-1}(q) - G^{-1}(q)$, enabling us to translate hypotheses about distributional comparisons into bounded treatment effect hypotheses.

Specifically, we will show that using the testing procedure in this paper we can test any hypothesis in the two-sample setting which can be written in terms of bounds, one- or two-sided, on the quantile difference function $\Delta(q)$. For example, the one-sided bounded quantile treatment effect hypothesis $\Delta(q) \leq \delta$ for all $q \in [0, 1]$ corresponds to the bounded treatment effect hypothesis $\tau_i \leq \delta$ for all i in our potential outcomes framework. Similarly, the two-sided bounded quantile treatment effect hypothesis $|\Delta(q)| \leq \delta$ for all $q \in [0, 1]$ corresponds to the bounded treatment effect hypothesis $|\tau_i| \leq \delta$ for all i.

4 A Sufficient Condition for Randomization-Style Tests

As is typical when constructing a hypothesis test, we will define a measurable test statistic $T : \mathcal{X} \to \mathbb{R}$ that takes as input the observed data X and outputs a real number. The purpose of the test statistic, as usual, is to impose an ordering on the sample space of observed data $\mathcal{X}$, where larger

values of the test statistic indicate stronger evidence against the null hypothesis $H_0 : P \in \mathcal{P}_0$. We know that we can construct an α-level test by rejecting the null hypothesis whenever the observed value of the test statistic exceeds the $(1-\alpha)$-quantile of its distribution under the null hypothesis, or equivalently when the probability of the test statistic under the null hypothesis exceeding the observed test statistic is less than α. Under the null, the probability of this happening is clearly less than α, so we know that the test is level-α i.e. rejects a true null hypothesis at most α proportion of the time.

A central challenge in nonparametric testing is that the distribution of the test statistic under the null hypothesis is often unknown or intractable. Randomization-style tests address this difficulty by exploiting settings in which the conditional distribution of the data, and hence of the test statistic, is known under the null, thereby enabling the construction of unconditionally valid tests from these conditional distributions.

The conditional distribution of the test statistic is then of central importance in testing, so I define it here.

Definition 1 (Conditional Distribution of the Test Statistic)**.** Let $T : \mathcal{X} \to \mathbb{R}$ be a test statistic, $\mathcal{S} \subseteq \mathcal{F}$ a sub-sigma algebra, and P a probability measure on $(\Omega, \mathcal{F})$. A *regular conditional distribution* of $T(X)$ given $\mathcal{S}$ under P is a function $\tilde{\kappa}_{P|\mathcal{S}} : \Omega \times \mathcal{B}(\mathbb{R}) \to [0,1]$ satisfying:

1. For each Borel set $B \subseteq \mathbb{R}$, the map $\omega \mapsto \tilde{\kappa}_{P|\mathcal{S}}(\omega, B)$ is $\mathcal{S}$-measurable and satisfies

$$\tilde{\kappa}_{P|\mathcal{S}}(\omega, B) = P\big(T(X) \in B \mid \mathcal{S}\big)(\omega) \quad \text{for } P\text{-a.e. } \omega.$$

2. For P-almost every ω, the map $B \mapsto \tilde{\kappa}_{P|\mathcal{S}}(\omega, B)$ is a probability measure on $\mathbb{R}$.

Informally, $\tilde{\kappa}_{P|\mathcal{S}}(\omega, B)$ gives the probability that $T(X)$ falls in B, conditional on the information in $\mathcal{S}$.

The existence of such a regular conditional distribution that can be written as a kernel is guaranteed by $T(X)$ being a measurable function from $(\mathcal{X}, \mathcal{F}^X)$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and standard results on regular conditional distributions; see e.g. Durrett (2019, Theorem 4.1.17).

Different versions of regular conditional distributions can be defined that are consistent with the above definition, which are formally equivalent up to P-null sets; that is, any two versions agree P-almost surely. For the purposes of the results that follow, we may therefore work with any arbitrary version of the regular conditional distribution without affecting our conclusions.

In the work that follows we are only interested in the conditional distribution of the test statistic given an information set that can be generated by a function of the observable data, or equivalently conditioning on a sub-sigma algebra $\mathcal{S} \subseteq \sigma(X)$ that is contained in the sigma-algebra generated by the data. The conditional probability is a random variable on the canonical space $(\Omega, \mathcal{F}, P)$, but since we are conditioning on a sub-sigma algebra of the data sigma-algebra, we can express it as a function of the observed data.

Definition 2 (Conditional Distribution of the Test Statistic on Observable Data)**.** Let $T : \mathcal{X} \to \mathbb{R}$ be a test statistic, $\mathcal{S} \subseteq \sigma(X)$ a sub-sigma algebra of that generated by the observable data X, and P a probability measure on $(\Omega, \mathcal{F})$ with P^X the pushforward probability measure on $(\mathcal{X}, \mathcal{F}^X)$. A *version of the conditional distribution of* $T(X)$ *given* $\mathcal{S}$ *on observable data* under P is a function $\kappa_{P|\mathcal{S}} : \mathcal{X} \times \mathcal{B}(\mathbb{R}) \to [0, 1]$ satisfying:

1. For each Borel set B, the map $x \mapsto \kappa_{P|\mathcal{S}}(x, B)$ is $\mathcal{S}$-measurable and satisfies

$$\kappa_{P|\mathcal{S}}(X(\omega), B) = P\big(T(X) \in B \mid \mathcal{S}\big)(\omega) \quad \text{for } P\text{-a.e. } \omega.$$

2. For P^X-almost every x, the map $B \mapsto \kappa_{P|\mathcal{S}}(x, B)$ is a probability measure on $\mathbb{R}$.

Informally, $\kappa_{P|\mathcal{S}}(x, B)$ gives the probability that $T(X)$ falls in B, conditional on the information in $\mathcal{S}$, evaluated at the observed data $X = x$.

Our goal is to compute the probability that the test statistic exceeds its observed value under the null hypothesis. To this end, we define the conditional tail probability as the conditional probability that the test statistic exceeds a given threshold r.

Definition 3 (Conditional Tail Probability)**.** Let $T : \mathcal{X} \to \mathbb{R}$ be a test statistic, $\mathcal{S} \subseteq \sigma(X)$ a sub-sigma algebra, P a probability measure on $(\Omega, \mathcal{F})$, and $\kappa_{P|\mathcal{S}}$ a version of the conditional distribution of $T(X)$ given $\mathcal{S}$ under P as defined above. The *conditional tail probability* function is defined as

$$\bar{F}_{P|\mathcal{S}}(x, r) := \kappa_{P|\mathcal{S}}(x, [r, \infty)) = \int_{\mathbb{R}} \mathbf{1}\{t \geq r\} \, \kappa_{P|\mathcal{S}}(x, dt)$$

for P^X-almost every $x \in \mathcal{X}$ and every $r \in \mathbb{R}$.

Informally, for a fixed threshold r and P-a.e. ω with $x = X(\omega)$, we have $\bar{F}_{P|\mathcal{S}}(x, r) = P(T(X) \geq r \mid \mathcal{S})(\omega)$; that is, the conditional tail probability gives the probability that the test statistic exceeds r, conditional on the information in $\mathcal{S}$.

When the null hypothesis is sharp, the conditional distribution of the data and thus test statistic are known under the null hypothesis. This allows us to compute the conditional tail probability exactly, and use it to construct valid randomization tests. Testing non-sharp null hypotheses, however, complicates this procedure since the conditional distribution of the data and test statistic are not known under the null hypothesis. The insight of this paper is that we can still construct valid randomization-style tests for non-sharp null hypotheses if we can find a known function that upper bounds the conditional tail probability under the null hypothesis. This is the content of Theorem 1 below.

In order to prove this theorem, we require the result that the conditional p-value, created by evaluating the conditional tail probability at the observed test statistic, is super-uniformly distributed for a given distribution P. This is stated formally in the lemma below, but first we define the conditional p-value.

Definition 4 (Conditional p-Value)**.** Let $T : \mathcal{X} \to \mathbb{R}$ be a test statistic, $\mathcal{S} \subseteq \sigma(X)$ a sub-sigma algebra, and P a probability measure on $(\Omega, \mathcal{F})$. The *conditional p-value* is defined by

$$p_{P|\mathcal{S}}(X) := \bar{F}_{P|\mathcal{S}}(X, T(X)) = \int_{\mathbb{R}} \mathbf{1}\{t \geq T(X)\}\, \kappa_{P|\mathcal{S}}(X, dt).$$

Lemma 1 (Super-Uniformity of Conditional p-Value)**.** *Let $T : \mathcal{X} \to \mathbb{R}$ be a test statistic, $\mathcal{S} \subseteq \sigma(X)$ a sub-sigma algebra, and P a probability measure on $(\Omega, \mathcal{F})$. Let $p_{P|\mathcal{S}}(X)$ denote the conditional p-value as defined in Definition 4. Then, for every $\alpha \in [0, 1]$,*

$$P\left(p_{P|\mathcal{S}}(X) \leq \alpha \,\middle|\, \mathcal{S}\right) \leq \alpha \qquad P\text{-a.s.}$$

The proof of this lemma can be found in the Appendix, Section A.1.

With this result in hand, we can now state and prove the main theorem of this section, which provides a sufficient condition for constructing valid randomization-style tests for non-sharp null hypotheses.

Theorem 1 (Sufficient Condition for a Valid Randomization Test)**.** *Let $H_0 : P \in \mathcal{P}_0$ be a null hypothesis, $T : \mathcal{X} \to \mathbb{R}$ a test statistic, and $\mathcal{S} \subseteq \mathcal{F}^X$ a sub-σ-algebra. For $P \in \mathcal{P}_0$ and $r \in \mathbb{R}$, let $\bar{F}_{P|\mathcal{S}}(\cdot, r)$ denote a version of the conditional tail probability $x \mapsto P\big(T(X) \geq r \mid \mathcal{S}\big)(x)$ under $X \sim P$.*

If there exists a known measurable function $U : \mathcal{X} \times \mathbb{R} \to [0, 1]$ such that for every $P \in \mathcal{P}_0$ and $r \in \mathbb{R}$,

$$U(x, r) \;\geq\; \bar{F}_{P|\mathcal{S}}(x, r) \qquad \text{for } P\text{-a.e. } x,$$

then $\phi(x) = \mathbb{1}\{\, U(x, T(x)) \leq \alpha \,\}$ is a level-α test of H_0.

Proof. Fix any $P \in \mathcal{P}_0$. By definition of ϕ,

$$E_P[\phi(X)] \;=\; E_P[\mathbb{1}\{\, U(X, T(X)) \leq \alpha \,\}]\,.$$

By the assumed upper bound $U(x, r) \geq \bar{F}_{P|\mathcal{S}}(x, r)$ (for P-a.e. x and all r), we have in particular

$$U(X, T(X)) \;\geq\; \bar{F}_{P|\mathcal{S}}(X, T(X)) \;=\; p_{P|\mathcal{S}}(X) \quad P\text{-a.s.}$$

where $p_{P|\mathcal{S}}(X)$ is the conditional p-value from Definition 4. Hence

$$\mathbb{1}\{\, U(X, T(X)) \leq \alpha \,\} \;\leq\; \mathbb{1}\{\, p_{P|\mathcal{S}}(X) \leq \alpha \,\} \quad P\text{-a.s.}. \tag{1}$$

Then it follows that

$$
\begin{aligned}
E_P[\phi(X)] &= E_P[\mathbb{1}\{\, U(X,T(X)) \leq \alpha \,\}] && \text{by definition of } \phi \\
&\leq E_P\big[\mathbb{1}\{\, p_{P|\mathcal{S}}(X) \leq \alpha \,\}\big] && \text{by (1)} \\
&= E_P\big[E_P\big[\mathbb{1}\{\, p_{P|\mathcal{S}}(X) \leq \alpha \,\} \mid \mathcal{S}\big]\big] && \text{by the law of total expectation} \\
&= E_P\big[P\big(p_{P|\mathcal{S}}(X) \leq \alpha \mid \mathcal{S}\big)\big] && \text{by definition of conditional probability} \\
&\leq E_P[\alpha] && \text{by Lemma 1} \\
&= \alpha.
\end{aligned}
$$

Therefore ϕ is a level-α test of H_0. □

The intuition behind Theorem 1 is that if we can find a known function that upper bounds the conditional tail probability of the test statistic under the null hypothesis, then we can use this upper bound to construct a valid test. The proof follows from the super-uniformity of the conditional p-value, which ensures that the probability of the conditional p-value being less than or equal to α is at most α. Since our upper bound U is always greater than or equal to the conditional p-value under the null, the event that $U(X,T(X)) \leq \alpha$ is a subset of the event that $p_{P|\mathcal{S}}(X) \leq \alpha$. Therefore, the probability of rejecting the null hypothesis using our test statistic is also bounded by α, ensuring that our test is level-α.

Essentially, Theorem 1 provides a general framework for constructing valid randomization-style tests for non-sharp null hypotheses whenever it is possible to find a known upper bound on the conditional tail probability of the test statistic under the null hypothesis. This insight opens up new possibilities for hypothesis testing in settings where traditional randomization tests are not applicable due to the non-sharp nature of the null hypothesis. It's worth noting that in this paper I do not exploit the full generality of applications of Theorem 1, and it presents a promising topic for future research: exploring for what other non-sharp null hypotheses there are known upper bounds on the conditional tail probability of certain classes of test statistics.

In this paper, we will apply Theorem 1 to develop finite-sample valid tests for bounded treatment effect null hypotheses. We will make use of three key components in order to generate U: First, one quality of the bounded treatment effect null hypothesis is that it puts restrictions on every realized set of potential outcomes under the null, unlike other non-sharp nulls such as those that restrict means or other moments of the treatment effect distribution which may only place restrictions on the probability of drawing a given "extreme" sample of potential outcomes. This means that without considering the actual distribution over the potential ouctomes, we can find a degenerate distribution over the potential outcomes that is "most extreme" in the sense of the size of treatment effects, compared to all other distributions satisfying the bounded treatment effect null. Second, we choose test statistics that are monotonic in the treatment effects so that we know this degenerate distribution will maximize the value of the test statistic among all distributions satisfying the bounded treatment effect null. Third, we rely on a known treatment assignment mechanism that

is independent of the potential outcomes, so that we can compute the conditional distribution of the test statistic under this degenerate distribution. By combining these three components, we can construct the upper bound U required by Theorem 1 and thus develop valid randomization-style tests for bounded treatment effect null hypotheses.

Before we continue to the specific bounded treatment effect null hypotheses, I will first connect the sufficient condition in Theorem 1 to the mainstream randomization test literature which conceives of these tests as being made possible by a group invariance condition.

4.1 Group Invariance as Sufficient Condition for Theorem 1

A broad strand of the literature, exemplified by Ritzwoller et al. (2025); Romano (1990); Chung and Romano (2013) and Lehmann and Romano (2022), conceptualizes randomization tests as being valid under a group invariance condition sometimes called the "randomization hypothesis" (Ritzwoller et al., 2025). This condition states that there are an algebraic group of transformations acting on the sample space of the observed data $\mathcal{X}$ such that the distribution of the observed data is invariant under these transformations when the null hypothesis is true. Formally, this condition can be stated as follows:

Condition 1 (Group Invariance Condition/Randomization Hypothesis)**.** Let G be a group of measurable transformations $g : \mathcal{X} \to \mathcal{X}$ with group operation given by composition of functions. Let $H_0 : P \in \mathcal{P}_0$ be the null hypothesis. The group invariance condition states that for every $P \in \mathcal{P}_0$ and every $g \in G$, the distribution of the transformed data $g(X)$ is the same as that of the original data X under P; that is, $P(g(X) = x) = P(X = x), P$-a.s..

When the group invariance condition holds, we can construct valid randomization tests by computing the distribution of the test statistic under the group of transformations G and comparing to the obesrved test statistic. This is equivalent to computing the conditional distribution of the test statistic given the orbit of the observed data under the group which is defined as the set $\{g(X) : g \in G\}$.

I have chosen not to define randomization tests in this way for two reasons. First, group invariance is a convenient sufficient condition, but many common designs (unequal-probability assignment, rerandomization/constrained randomization, and covariate-adaptive or sequential procedures) where tests of a similar form to randomization tests are suitable, do not admit a natural uniform group-orbit representation. In these cases, valid randomization-style inference instead follows from reproducing the known assignment mechanism (possibly after conditioning on design-relevant statistics) to generate the reference distribution (Branson and Bind, 2019; Morgan and Rubin, 2015; Bugni et al., 2018). See also Zhang and Zhao (2023) for a complementary discussion emphasizing the broadness of randomization-like testing procedures beyond the group invariance condition.

Second, when expanding randomization tests to non-sharp null hypotheses, it is not always clear how to define a group invariance condition that holds under the null hypothesis. By defining

randomization tests in terms of known tail distributions or upper bounds, we can avoid this difficulty and still construct valid tests for non-sharp null hypotheses.

To see explicitly how the group invariance condition implies the sufficient condition in Theorem 1, I state the following Proposition:

Proposition 1 (Group invariance is sufficient for Theorem 1)**.** *Suppose Condition 1 holds for a null hypothesis $H_0 : P \in \mathcal{P}_0$ with a* finite *group G of measurable transformations acting on $\mathcal{X}$. Define the (set-valued) orbit map*

$$\text{orb}(x) := \{g(x) : g \in G\} \subseteq \mathcal{X},$$

and let $\mathcal{S} := \sigma(\text{orb}(X))$ denote the σ-algebra generated by the orbit of the realized data.

Then the function

$$U(x, r) \;:=\; \frac{1}{|G|} \sum_{g \in G} \mathbb{1}\{\, T(g(x)) \geq r \,\}, \qquad (x, r) \in \mathcal{X} \times \mathbb{R},$$

is known and measurable, and satisfies, for every $P \in \mathcal{P}_0$ and $r \in \mathbb{R}$,

$$U(x, r) \;=\; \bar{F}_{P|\mathcal{S}}(x, r) \qquad \textit{for } P^X\textit{-a.e. } x.$$

This result follows from the fact that under the group invariance condition, the conditional distribution of the test statistic given the orbit of the data is uniform over the values of the test statistic computed on each element of the orbit. To see this, consider any two members of an orbit, say $x_1 = g_1(x)$ and $x_2 = g_2(x)$ for some $g_1, g_2 \in G$. By the group invariance condition, the distribution of the data under the null hypothesis is invariant under the transformations in G, so we have $P(X = x_1) = P(X = x_2)$. This implies that the conditional distribution of the test statistic given the orbit is uniform over the values of the test statistic computed on each element of the orbit. Therefore, we can compute the conditional tail probability as proportion of elements in the orbit for which the test statistic exceeds a given threshold r, which is exactly the function $U(x, r)$ defined in the proposition.

As the upper bound holds with equality under the group invariance condition, Theorem 1 implies that the randomization test that rejects the null hypothesis whenever $U(X, T(X)) \leq \alpha$ is not only a valid level-α test but has size of exactly α.

5 One-Sided Bounded Treatment Effect Nulls

There are many situations in which researchers want to test one-sided bounds on individual treatment effects. For instance, policymakers may wish to verify that a new intervention does not harm any treated individual, or at least that any harm is smaller than an acceptable margin δ (e.g. $\tau_i \geq -\delta$ for all i). Similar questions are common in clinical trials, where the goal is to rule out clinically meaningful deterioration relative to a standard of care. In pilot programs, administrators

may instead be interested in whether the program generates a meaningfully positive impact for at least some individuals (or for at least one pre-specified subgroup), for example whether there exists any i such that $\tau_i \geq \delta$ for a practically relevant threshold $\delta > 0$.

A related example, mentioned earlier, is the monotonicity assumption in instrumental-variables designs. Monotonicity can be written as a one-sided bound on unit-level changes in treatment status induced by the instrument: $\tau_i^D = D_i(1) - D_i(0) \geq 0$, which rules out "defiers." Testing this hypothesis, together with the instrument assignment mechanism, can serve as a useful specification check for an IV design.

Finally, after drawing the analogy between individual treatment effects in a binary-treatment potential-outcomes framework and quantile differences in the two-sample problem, a broader set of applications becomes available. Many distributional comparisons can be expressed as one-sided bounds on quantile differences. For example, first-order stochastic dominance of distribution 1 over distribution 0 is equivalent to $Q_1(p) - Q_0(p) \geq 0$ for all $p \in (0, 1)$ (or, equivalently, $F_1(y) \leq F_0(y)$ for all y). Variants with a tolerance margin δ correspond to requiring $Q_1(p) - Q_0(p) \geq -\delta$ uniformly over p.

The testing process outlined in this section yields tests for these hypotheses that are finite-sample valid under known treatment assignment mechanisms, in the sense of controlling size at (or below) the nominal level α.

I formalize one-sided bounded treatment effect hypotheses by allowing the bound to vary across units as a *known* function of observable characteristics and (at most) one potential outcome. Let

$$\tau_i \;:=\; Y_i(1) - Y_i(0)$$

denote the individual treatment effect, and let Z_i collect observed covariates (or other pre-treatment characteristics).

Definition 5 (One-sided lower bounded treatment effect null)**.** Fix a known measurable function $l : \mathbb{R} \times \mathcal{Z} \to \mathbb{R}$. The *one-sided lower bounded treatment effect null* is

$$H_0^{\ell} :\; P\Big(\tau_i \geq l\big(Y_i(0), Z_i\big) \text{ for all } i\Big) = 1.$$

Equivalently, H_0^{ℓ} imposes the pointwise restriction

$$Y_i(1) \;\geq\; Y_i(0) + l\big(Y_i(0), Z_i\big) \qquad \text{for all } i$$

almost surely under P.

Definition 6 (One-sided upper bounded treatment effect null)**.** Fix a known measurable function $u : \mathbb{R} \times \mathcal{Z} \to \mathbb{R}$. The *one-sided upper bounded treatment effect null* is

$$H_0^{u} :\; P\Big(\tau_i \leq u\big(Y_i(0), Z_i\big) \text{ for all } i\Big) = 1,$$

or equivalently,

$$Y_i(1) \;\leq\; Y_i(0) + u\big(Y_i(0), Z_i\big) \qquad \text{for all } i$$

almost surely under P.

Any bounded treatment null hypothesis is characterized by either a lower bounding function $l(Y_i(0), Z_i)$ or an upper bounding function $u(Y_i(0), Z_i)$. Allowing the bound to depend on Z_i accommodates heterogeneous bounds across observable subpopulations. This permits testing hypotheses of maximal harm or minimal benefit for different popoulations, including targeting populations at-risk or populations for whom the cost of treatment may vary. For example, in a clinical trial, the acceptable harm threshold for average heart rate may differ between older and younger patients due to varying risk profiles, so we may choose a lower bounding function $l(y, z)$ that depends on age group z:

$$l(y, z) = \begin{cases} -0.5, & \text{if } z > 50, \\ -1, & \text{if } z \leq 50 \ . \end{cases}$$

Testing the resulting null hypothesis H_0^ℓ then verifies that no patient experiences a heart rate increase exceeding the age-specific threshold.

Many econometric applications may be concerned with testing the impact of interventions that vary in cost across units. Consider the setting studied by Evans et al. (2021), who evaluate the impacts on costly hospital readmission rates from a social-worker lead transition program for patients after being discharged from the hospital. The cost of providing the social worker intervention varies across patients according to observable pre-treatment characteristics such as the severity of their illness, social support networks and geographic location. A policy-maker may wish to test whether the intervention reduces readmission rates by an amount that justifies the cost of providing the intervention, which varies across patients. This can be expressed as a one-sided bounded treatment effect null hypothesis with a lower bounding function $l(y, z)$ that depends on patient characteristics, like z, the distance from the patient's home to the nearest hospital (which affects the cost of providing the intervention). For example, we can imagine a bounding function of the form

$$l(y, z) = \begin{cases} -0.1, & \text{if } z \leq 5 \text{ miles}, \\ -0.2, & \text{if } 5 < z \leq 15 \text{ miles}, \\ -0.3, & \text{if } z > 15 \text{ miles} \ . \end{cases}$$

Testing the null hypothesis H_0^ℓ then verifies that the program reduces future hospital costs by at least as much as the expected cost of the intervention. Alternatively, by setting a very similar upper bounding function, a policy-maker may test whether the intervention decreases future hospital costs by more than a cost-justifying threshold for any patient.

In the definitions above, I write the upper and lower bounding functions in terms of the untreated potential outcome $Y_i(0)$, however this procedure works equally well if we were to write these functions in terms of the treated potential outcome. I choose to write in terms of the untreated

potential outcome because it leads to more natural, straightforward interpretations of the bounded hypothesis.

It is worth noting why we cannot write the bounding functions in terms of both potential outcomes simultaneously. The bounded treatment effect null hypothesis restricts the set of potential outcome schedules that can be realized with positive probability. If the bounding function were to depend on both $Y_i(0)$ and $Y_i(1)$, then knowing the bound would require knowing both potential outcomes for each unit. If only one potential outcome were known, the resaercher could derive a bound by taking the supremum or infinum over the function in the unknown dimension. This supremum or infinum value can be interpreted as a bounding function itself (i.e. $u(Y_i(0), Z_i) = \sup_{y \in \mathcal{Y}_1}\{u(Y_i(0), y, Z_i))\}$ and holds all relevant bounding information from originally defined function. For this reason, I restrict the bounding functions to depend on at most one potential outcome.

It may seem counterintuitive that we can specify bounds on the treatment effect that depend on an unobserved potential outcome, but this is no different from other non-sharp null hypotheses in the potential outcomes framework (e.g., nulls on average treatment effects or quantile treatment effects) that also depend on unobserved potential outcomes. The testing procedure will consider the least favourable potential outcome schedule that is consistent with this bounding function, so the test remains valid even though the bounding function depends on unobserved potential outcomes.

Dependence on $Y_i(0)$ permits restrictions on the treatment effect in a manner that is dependent on the untreated counterfactual, which fits settings where the researcher may be concerned with the proportional size of treatment effects or where value of the outcome is non-linear. One example of an economic application in which a "meaningful" effect is naturally stated in proportional terms is savings balances, an outcome which is strictly positive, highly skewed, and varies widely in scale across individuals. For example, Ashraf et al. (2006) study a randomized offer of a commitment savings product and report impacts in percentage terms (e.g., large increases in savings balances relative to a control group). Such settings motivate bounded-effect hypothesis that tests whether the savings intervention increases balances by at least a proportion $\rho > 0$ for all individuals, by setting a lower bounding function:

$$l(y, z) = \rho y.$$

Or alternatively, providing evidence for the alternative hypothesis that the intervention is impactful at or above this threshold for some component of the population, which can be done by testing the upper-bounded hypothesis indexed by $u(y, z) = \rho y$. Unlike additive bounds $\tau_i \geq \delta$, proportional bounds scale with the untreated counterfactual and provide a common notion of "practically meaningful" improvement across individuals with different baseline levels.

Taking the upper or lower bounding function to be a constant delivers several familiar cases mentioned above, such as the no-harm null $l(y, z) \equiv 0$, and margin nulls $l(y, z) \equiv \delta$ or $u(y, z) \equiv \delta$ for a fixed $\delta \in \mathbb{R}$.

What makes testing these one-sided bounded treatment effect nulls possible in finite sample is that we can choose test statistics for which computing the tail probability under a potential

outcome schedule that attains the bound (that is, where $\tau_i = l(Y_i(0), Z_i)$ or $\tau_i = u(Y_i(0), Z_i)$ for all i) provides an upper bound on the tail probability under any distribution over potential outcomes satisfying the null. This family of test statistics is monotonic in the individual treatment effects, meaning that increasing (or decreasing) the treatment effect for any unit cannot decrease (or increase) the value of the test statistic. It is intuitive then that if we choose a potential outcome schedule that has the highest or lowest possible treatment effects allowed under the null, then for each assignment of treatment, the value of the test statistic will be maximized (or minimized) compared to all other potential outcome distributions with the same treatment assignment.

We call this family of test statistics *effect-increasing* following Caughey et al. (2023) and Rosenbaum (2002), and define them formally below.

Definition 7 (Effect-increasing test statistic)**.** A statistic $T : \{0,1\}^n \times \mathbb{R}^n \to \mathbb{R}$ is *effect-increasing* if for all treatment-assignment vectors $d \in \{0,1\}^n$, all outcome vectors $y \in \mathbb{R}^n$, and all vectors $r, \zeta \in \mathbb{R}^n_{\geq 0}$,

$$T\Big(d,\ y + r \circ d - \zeta \circ (\mathbb{1} - d), z\Big)\ \geq\ T(d, y, z),$$

where $\circ$ denotes the Hadamard (componentwise) product and $\mathbb{1}$ is the n-vector of ones.

Intuitively, we can understand that holding the assignment d fixed, an effect-increasing statistic is weakly increasing in the outcomes of treated units and weakly decreasing in the outcomes of untreated units. Equivalently, increasing any individual treatment effect (by raising a treated outcome and/or lowering an untreated outcome) cannot decrease the value of T.

Theorem 2 formalizes the fact that by choosing an effect-increasing test statistic, when we are in a setting where the treatment assignment mechanism is known (up to conditioning on a sub-σ-algebra $\mathcal{S}$), we can construct a valid level-α test for one-sided bounded treatment effect null hypotheses by computing the tail probability of the test statistic under a potential outcome schedule that attains the bound.

Theorem 2 (Valid Test for One-Sided Bounded Treatment Effects)**.** *Let $H_0^\ell : P \in \mathcal{P}_0^\ell$ be the one-sided lower bounded treatment effect null as in Definition 5, or let $H_0^u : P \in \mathcal{P}_0^u$ be the one-sided upper bounded treatment effect null as in Definition 6. Suppose the conditional distribution of treatment $P(D = d \mid \mathcal{S})$ is known and independent of potential outcomes, where $\mathcal{S}$ is a sub-σ-algebra of the data σ-algebra. Let T be an* effect-increasing *test statistic (Definition 7).*

(i) Upper bounded null H_0^u: *Define*

$$U_u(x, r) = \sum_{\tilde{d} \in \mathrm{Supp}\{D\}} \mathbb{1}\Big\{T\big(\tilde{d}, \tilde{y}^u, z\big) \geq r\Big\} \cdot P\big(D = \tilde{d} \mid \mathcal{S}\big),$$

$$\tilde{y}_i^u = \begin{cases} y_i + u(y_i, z_i) & \text{if } d_i = 0, \tilde{d}_i = 1, \\ \inf\{y : y + u(y, z_i) \geq y_i\} & \text{if } d_i = 1, \tilde{d}_i = 0, \\ y_i & \text{if } \tilde{d}_i = d_i. \end{cases}$$

Then $\phi_u(x) = \mathbb{1}\{U_u(x, T(x)) \leq \alpha\}$ is a valid level-α test of H_0^u.

(ii) Lower bounded null H_0^ℓ: *Define the* effect-decreasing *statistic* $T^-(d, y, z) := -T(d, y, z)$, *and*

$$U_\ell(x, r) = \sum_{\tilde{d} \in \text{Supp}\{D\}} \mathbb{1}\Big\{T^-\big(\tilde{d}, \tilde{y}^\ell, z\big) \geq r\Big\} \cdot P\big(D = \tilde{d} \mid \mathcal{S}\big),$$

$$\tilde{y}_i^\ell = \begin{cases} y_i + l(y_i, z_i) & \text{if } d_i = 0, \tilde{d}_i = 1, \\ \sup\{y : y + l(y, z_i) \geq y_i\} & \text{if } d_i = 1, \tilde{d}_i = 0, \\ y_i & \text{if } \tilde{d}_i = d_i. \end{cases}$$

Then $\phi_\ell(x) = \mathbb{1}\{U_\ell(x, T^-(x)) \leq \alpha\}$ *is a valid level-*α *test of* H_0^ℓ.

The proof of this theorem is included in the appendix, in Section A.3.

Theorem 2 provides a concrete construction of the upper bound function U required by Theorem 1 for testing one-sided bounded treatment effect null hypotheses. The key insight is that when treatment effects are bounded above by $u(Y_i(0), Z_i)$ or below by $l(Y_i(0), Z_i)$, we can construct a "worst-case" potential outcome schedule where each unit's treatment effect attains exactly this bound. Given an effect-increasing test statistic (or its negation for lower bounds), this worst-case schedule maximizes the test statistic among all distributions satisfying the null, and thus the tail probability computed under this schedule provides an upper bound on the tail probability under any null-consistent distribution.

For the upper bounded null H_0^u, the function $U_u(x, r)$ computes this upper-bounding tail probability by: (1) considering all possible treatment assignments $\tilde{d}$ in the support of the assignment mechanism; (2) for each such assignment, imputing what the observed outcomes $\tilde{y}^u$ would be if treatment effects were exactly at their upper bounds; and (3) summing the known assignment probabilities over assignments where test statistic exceeds r. The imputation rule for $\tilde{y}_i^u$ reconstructs what the outcome would be under assignment $\tilde{d}_i$ given that we observe y_i under d_i: if a control unit becomes treated, we add the maximal effect $u(y_i, z_i)$; if a treated unit becomes control, we invert the bound to recover the untreated potential outcome that maximizes the treatment effect; if assignment is unchanged, the outcome remains the same.

For the lower bounded null H_0^ℓ, we apply the same logic but with the effect-decreasing statistic $T^- = -T$. Large values of T^- correspond to small values of T, which occur when treatment effects are small. The worst case under a lower bound is thus when effects are exactly at their lower bounds $l(Y_i(0), Z_i)$. The function $U_\ell(x, r)$ computes the tail probability of T^- under this worst-case schedule analogously to U_u.

5.1 Application to Two-Sample Problem

Recall from Section 3 that the two-sample problem can be embedded in the potential outcomes framework such that bounds on the treatment effect correspond to bounds on the quantile differ-

ences between the two distributions. Specifically,

$$\tau_i = Y_i(1) - Y_i(0)$$

corresponds to the quantile difference

$$F^{-1}(q_i) - G^{-1}(q_i),$$

where q_i is the latent quantile associated with unit i.

Thus, bounded treatment effect null hypotheses of the form

$$P\big(\tau_i \leq u(Y_i(0), Z_i) \text{ for all } i\big) = 1$$

correspond to restrictions on the quantile treatment effect function. In particular, when the bound u depends only on the control potential outcome, we have

$$F^{-1}(q) - G^{-1}(q) \leq u(G^{-1}(q)) \quad \text{for all } q \in [0,1].$$

This connection yields the following corollary of Theorem 2:

Corollary 1 (Valid Tests for Bounded Quantile Treatment Effects in Two-Sample Problems)**.** *Consider the two-sample setting with $X_i \overset{iid}{\sim} F$ for $i = 1, \ldots, n_X$ and $Y_j \overset{iid}{\sim} G$ for $j = 1, \ldots, n_Y$. Let T be an effect-increasing test statistic.*

(i) *The test ϕ_u from Theorem 2(i) with $u(y, z) = u(y)$ is a valid level-α test of*

$$H_0^u : F^{-1}(q) - G^{-1}(q) \leq u(G^{-1}(q)) \quad \textit{for all } q \in [0,1].$$

(ii) *The test ϕ_ℓ from Theorem 2(ii) with $l(y, z) = l(y)$ is a valid level-α test of*

$$H_0^\ell : F^{-1}(q) - G^{-1}(q) \geq l(G^{-1}(q)) \quad \textit{for all } q \in [0,1].$$

Proof. The result follows immediately from Theorem 2 after embedding the two-sample problem in the potential outcomes framework with the latent quantile construction. □

Corollary 1 provides finite-sample valid tests for several important distributional hypotheses:

Setting $u(y) \equiv 0$ in Corollary 1(i) yields a test of first order stochastic dominance (FOSD),

$$H_0^{\text{FOSD}} : F^{-1}(q) - G^{-1}(q) \leq 0 \quad \text{for all } q \in [0,1],$$

equivalently $F(y) \geq G(y)$ for all y, meaning G first-order stochastically dominates F. Under H_0^{FOSD}, the least favorable configuration is $\tau_i = 0$ for all i, corresponding to equality of distributions ($F = G$). This sharp null renders the pooled sample exchangeable, so ϕ_u reduces to an ordinary permutation test. For stochastic dominance alternatives, a natural choice is the one-sided

Kolmogorov–Smirnov statistic $T_{\mathrm{KS}} = \sup_y(\hat{G}(y) - \hat{F}(y))$, though any effect-increasing statistic yields the same finite-sample validity. This result confirms and generalizes a finding in Bugni et al. (2025), who establish validity for the Kolmogorov–Smirnov statistic; my framework extends this to any effect-increasing test statistic.

Setting $u(y) \equiv \delta$ for a fixed margin $\delta > 0$ tests

$$H_0^{\delta\text{-FOSD}} : F^{-1}(q) - G^{-1}(q) \leq \delta \quad \text{for all } q \in [0,1],$$

permitting violations of dominance up to a constant tolerance. The least favorable null is $\tau_i = \delta$ for all i, which can be reduced to exchangeability by shifting treated observations: define $X_i^{(\delta)} = X_i - \delta$ and permute treatment labels over the pooled transformed sample. Rejecting $H_0^{\delta\text{-FOSD}}$ provides finite-sample evidence that the quantile treatment effect exceeds δ somewhere. Inverting this family of tests across δ yields a one-sided finite-sample upper confidence bound on $\sup_q\{F^{-1}(q) - G^{-1}(q)\}$.

5.2 Simulation Results for a Null of "No Harm"

To illustrate the finite-sample properties of the one-sided bounded treatment effect test developed in Theorem 2, I conduct a Monte Carlo simulation study. The simulations evaluate size control under the null hypothesis and power against various alternatives for the "no harm" hypothesis $H_0^{\ell} : \tau_i \geq 0$ for all i.

The simulation considers a completely randomized experiment with n units, half assigned to treatment. Control potential outcomes are drawn as $Y_i(0) \overset{iid}{\sim} N(0,1)$. I examine three sample sizes: $n \in \{14, 18, 22\}$, which are small enough to permit exact enumeration of all $\binom{n}{n/2}$ treatment assignments. For each scenario, I conduct 200-500 Monte Carlo replications and compute the rejection rate at significance level $\alpha = 0.05$.

The test follows Theorem 2(ii) for lower-bounded nulls. Under the boundary sharp null where $\tau_i = 0$ for all i, the test reduces to a standard permutation test using the negated difference-in-means as the test statistic. The p-value is computed as the proportion of permutations yielding a test statistic at least as extreme as the observed value.

I consider six treatment effect distributions spanning both null and alternative hypotheses:

Null configurations: (1) Sharp null boundary with $\tau_i = 0$ for all i; (2) Interior null with small positive effects, $\tau_i \overset{iid}{\sim}$ Uniform$(0.05, 0.15)$; (3) Interior null with large positive effects, $\tau_i \overset{iid}{\sim}$ Uniform$(0.2, 0.4)$.

Alternative configurations: (4) Weak alternative with $\tau_i \overset{iid}{\sim} N(0, 0.3)$, where approximately half of units experience harm; (5) Strong alternative with $\tau_i \overset{iid}{\sim} N(-0.2, 0.3)$, where the majority of units are harmed; (6) Mixed alternative where 70% of units have positive effects drawn from Uniform$(0.1, 0.3)$ and 30% have negative effects from Uniform$(-0.3, -0.1)$.

Table 1 reports rejection rates across all scenarios. Under all three null configurations, rejection rates remain at or below the nominal level of 0.05, confirming finite-sample size control as established in Theorem 2. The test exhibits slight conservatism for interior null configurations, which

Table 1: Rejection Rates for Lower-Bounded Treatment Effect Test

DGP	$n = 14$	$n = 18$	$n = 22$
Null: $\tau_i = 0$ (sharp)	0.042	0.048	0.045
Null: $\tau_i \sim U(0.05, 0.15)$	0.040	0.032	0.045
Null: $\tau_i \sim U(0.2, 0.4)$	0.016	0.014	0.005
Alternative: $\tau_i \sim N(0, 0.3)$	0.068	0.050	0.040
Alternative: $\tau_i \sim N(-0.2, 0.3)$	0.084	0.118	0.125
Alternative: 70% positive, 30% negative	0.034	0.036	0.050

Note: Rejection rates based on 500 simulations for $n \in \{14, 18\}$ and 200 simulations for $n = 22$. Testing $H_0 : \tau_i \geq 0$ for all i at $\alpha = 0.05$ level. Test uses difference-in-means statistic with full permutation distribution.

is expected since the randomization distribution is computed at the boundary where effects are exactly zero.

Against alternatives, the test demonstrates moderate power even in these small samples. The strong alternative yields rejection rates of 6–12%, while the weak and mixed alternatives show more modest power of 1–7%. The relatively low power reflects the stringency of the hypothesis: rejecting H_0^ℓ requires sufficient evidence that *at least one* unit is harmed, which demands substantial signal when sample sizes are limited. These results illustrate the finite-sample validity properties of the proposed test and suggest its practical applicability in settings where asymptotic approximations may be unreliable.

5.3 Example 1: Monotonicity of Instrumental Variable

We revisit the water connection experiment of Devoto et al. (2012), where a randomized encouragement instrument $R_i \in \{0, 1\}$ is used to study the effect of formal water connection $C_i \in \{0, 1\}$ on household outcomes. Let $C_i(r)$ denote the potential treatment (connection) status under instrument assignment r. The monotonicity assumption requires that encouragement does not reduce any household's propensity to connect:

$$H_0^{\text{mono}} : \quad P\big(C_i(1) - C_i(0) \geq 0 \text{ for all } i\big) = 1.$$

This is precisely a one-sided lower bounded treatment effect null with $l \equiv 0$, where the "treatment" is instrument assignment R_i and the "outcome" is connection status C_i. Theorem 2(ii) therefore provides a finite-sample valid test of monotonicity using any effect-increasing statistic computed on the observed (R_i, C_i) pairs, with the randomization distribution determined by the known block-level instrument assignment mechanism.

The test implements Theorem 2(ii) for the lower-bounded null $H_0^{\text{mono}} : \tau_i \geq 0$ for all i, where $\tau_i = C_i(1) - C_i(0)$ denotes the change in connection status induced by encouragement. The test statistic is the negated difference-in-means, $T^- = -(\bar{C}_1 - \bar{C}_0)$, which is effect-decreasing (equivalently, $-T^-$ is effect-increasing). Under the boundary sharp null $\tau_i = 0$ for all i, potential outcomes are fully

imputable as $C_i(0) = C_i(1) = C_i$ for all i, rendering the sample exchangeable within blocks.

For our testing and inference procedure, the conditioning σ-algebra $\mathcal{S}$ is generated by the event that a particular set of realized outcomes is observed. To derive the conditional distribution of the test statistic, we condition on $\mathcal{S}$ or equivalently on the event of having drawn a sample with a schedule of potential outcomes consistent with the observed sample. Given this conditioning, the remaining randomness in the data arises solely from the known assignment mechanism.

The treatment (instrument) assignment in Devoto et al. (2012) is conducted at the cluster level rather than the household level to avoid assigning immediate neighbors to different groups (since the intervention involves differential access to information). Clusters are defined by micro-geographic proximity: two adjacent plots, two plots facing each other, or plots up to one house apart are grouped into the same cluster. This produces 626 clusters from 732 plots containing 845 households. These clusters are randomly assigned to treatment (315 clusters, 434 households) or control (311 clusters, 411 households), with stratification by location, water source, number of children under five, and number of households within the cluster.

For each permuted assignment $\tilde{d}$ consistent with the stratified cluster-randomized design, we compute $T^{-}(\tilde{d}, C, z)$ under the imputed outcomes and compare to the observed statistic. With 10,000 Monte Carlo draws from the stratified permutation distribution, the p-value approximates $P(T^{-}_{\text{perm}} \geq T^{-}_{\text{obs}} \mid \mathcal{S})$. While I do not formally demonstrate how Monte Carlo approximation preserves finite-sample validity here, the argument is similar to the typical arguments in the literature. More formally, consider a random set defined on the same probability triple which indicates 10,000 independent draws from the permutation distribution of valid treatment assignments. Then augment the σ-algebra $\mathcal{S}$ with the event that D falls within this set. Then, the Monte Carlo test is simulating the conditional distribution of the data on this new σ-algebra and the test is valid in finite samples by Theorem 2.

The full sample comprises $n = 797$ households with an average of 1.33 households per block; the informal-connection subgroup has $n = 172$ across 138 blocks. Finite-sample validity follows from Theorem 2 because: (i) the block-stratified assignment is known by design; (ii) assignment is independent of potential outcomes conditional on blocks; and (iii) the negated difference-in-means satisfies the effect-monotonicity condition required for boundary completion.

Table 2: Monotonicity Tests for Devoto et al. (2012) IV Design

Sample	n	Blocks	Test Statistic	p-value
Full Sample	797	600	-0.602	1.000
Informal Connection	172	138	-0.548	1.000

Table 2 reports results from testing the monotonicity hypothesis $H_0^{\text{mono}} : P(C_i(1) - C_i(0) \geq 0 \text{ for all } i) = 1$ for the full sample and for households with informal water connections at baseline. P-values of 1.000 in both samples indicate no evidence against monotonicity: the data are fully consistent with the assumption that encouragement weakly increases (or leaves unchanged) every household's propensity to obtain a formal connection. This provides reassurance that the LATE

interpretation of the IV estimates in Devoto et al. (2012) is supported by the experimental evidence, including for the subgroup of households relying on informal connections where concerns about defiance might be most pronounced.

5.4 Example 2: Inference on Treatment Effect Bounds

The previous subsection demonstrated how to test the monotonicity assumption in an instrumental variables design by applying Theorem 2 to a "no defiers" null hypothesis. We now turn to a complementary application: constructing finite-sample valid confidence bounds on the worst-case individual-level treatment effect.

In the microcredit experiment of Banerjee et al. (2015), the policy question is not merely whether a "no harm" hypothesis can be rejected, but rather: *what is the largest individual-level harm we can rule out with high confidence?* Let $\tau_i = Y_i(1) - Y_i(0)$ denote the individual treatment effect on a household outcome of interest (e.g., consumption, profits, or children's time use). We seek a finite-sample valid lower confidence bound $\underline{\tau}_{1-\alpha}$ such that

$$P\Big(\tau_i \geq \underline{\tau}_{1-\alpha} \text{ for all } i\Big) \geq 1 - \alpha.$$

This bound is obtained by inverting the family of one-sided tests from Theorem 2(ii). For each candidate threshold $\delta \in \mathbb{R}$, we test the lower-bounded null $H_0^{\ell}(\delta) : \tau_i \geq \delta$ for all i. The $(1-\alpha)$-confidence lower bound is then

$$\underline{\tau}_{1-\alpha} := \sup\Big\{\delta \in \mathbb{R} : \phi_{\ell}^{\delta}(X) = 0\Big\},$$

and the corresponding $(1-\alpha)$-confidence interval for the minimum individual treatment effect is

$$CI_{1-\alpha} = \big(-\infty,\, \underline{\tau}_{1-\alpha}\big].$$

By construction, this confidence interval is always unbounded below. To see why, note that the one-sided test ϕ_{ℓ}^{δ} rejects when there is sufficient evidence that at least one unit has $\tau_i < \delta$. As $\delta \to -\infty$, the null hypothesis $H_0^{\ell}(\delta) : \tau_i \geq \delta$ for all i becomes increasingly weak, it merely asserts that no individual experiences an arbitrarily large negative effect. For any finite sample, there will always exist some sufficiently negative δ^* such that we fail to reject $H_0^{\ell}(\delta)$ for all $\delta \leq \delta^*$. Consequently, the set $\{\delta : \phi_{\ell}^{\delta}(X) = 0\}$ is always unbounded below, and the confidence interval extends to $-\infty$.

The experiment in Banerjee et al. (2015) uses a cluster-randomized matched-pair design. From an initial set of 120 eligible neighborhoods (slums/bastis) in Hyderabad, 16 were excluded for safety reasons, leaving 104 neighborhoods for the study. These 104 neighborhoods were grouped into 52 matched pairs based on baseline area characteristics, specifically, average per-capita consumption and average per-household debt. Within each matched pair, one neighborhood was randomly assigned to receive early access to Spandana microfinance operations (treatment), while the other served as a control with delayed program entry.

For our testing and inference procedure, as above, the conditioning σ-algebra $\mathcal{S}$ is generated by the event that a particular set of realized outcomes is observed. To derive the conditional distribution of the test statistic, we condition on $\mathcal{S}$ or equivalently on the event of having drawn a sample with a schedule of potential outcomes consistent with the observed sample. Given this conditioning, the remaining randomness in the data arises solely from the known assignment mechanism. Treatment is randomized within matched pairs: in each of the 52 pairs, exactly one neighborhood is assigned to treatment, so the support of the assignment distribution consists of 2^{52} possible allocations.

Under the boundary sharp null hypothesis $\tau_i = \delta$ for all i, the unobserved potential outcomes are imputed by a constant shift, setting $Y_i(0) = Y_i - \delta$ for treated units and $Y_i(1) = Y_i + \delta$ for control units. The p-value is then computed by permuting treatment assignment within each matched pair, flipping the treated and control neighborhoods while holding the pair structure fixed, and comparing the resulting permutation distribution of the test statistic to its observed value.

The test statistic is the difference-in-means, $T = \bar{Y}_{\text{treat}} - \bar{Y}_{\text{control}}$, computed at the household level; as we are testing the lower-bounded null $H_0^\ell : \tau_i \geq 0$, we use the effect-decreasing statistic $T^- = -T$. Since the exact permutation distribution over all 2^{52} within-pair reassignments is computationally infeasible, we approximate it using 5,000 Monte Carlo draws, each obtained by independently flipping treatment assignment within each matched pair. This Monte Carlo approximation inherits finite-sample validity because, under the null, each permuted assignment is equally likely.

Table 3: 95% Confidence Intervals for Lower Bound of Treatment Effects

Outcome	N	Control Mean	Treatment Mean	Diff in Means	p-value ($\ell = 0$)	95% CI for Lower Bound
Total monthly expenditure	6827	6375	6482	107	0.671	$(-\infty, 498]$
Nondurable expenditure	6781	5831	5848	16	0.536	$(-\infty, 352]$
Food expenditure	6827	2325	2276	−49	0.248	$(-\infty, 63]$
Health expenditure	6827	630	624	−6	0.463	$(-\infty, 88]$
Business profits	6239	745	1171	426	0.899	$(-\infty, 962]$

Notes: 95% confidence intervals computed by inverting finite-sample cluster-randomized permutation tests. CI contains all values ℓ where we fail to reject H_0: $\tau_i \geq \ell$ at $\alpha = 0.05$. Data from Banerjee et al. (2015), Endline 1 (15–18 months post-treatment). 104 neighborhoods (52 treatment, 52 control). All values in rupees per month.

Table 3 reports 95% confidence intervals for the lower bound of individual-level treatment effects across six household outcomes from the Hyderabad microcredit experiment.

The results provide finite-sample valid bounds on the worst-case harm from microcredit access. For all expendeiture categories, the hyopthesis test of no-harm ($\ell = 0$) fails to reject, indicating no substantial evidence of negative effects for a given section of the population.

For all expenditure categories the 95% confidence intervals are bounded above by a positive number meaning that we cannot reject the hypothesis that microcredit access leasd to higher

expenditure and consumption in all considered categories. For instance, for total monthly expenditure, we can only rule out individual-level treatment effects exceeding expenditure increases of 498 rupees with 95% confidence. We can conlude then that our analysis provides no compelling evidence for harm to long-term consumption due to debt repayment demands from microcredit access.

6 Two-Sided Bounded Treatment Effect Nulls

In many applications, the relevant scientific question is not whether treatment helps or harms, but whether it can move outcomes in either direction beyond a certain threshold. This is common in "equivalence" or "no relevant difference" settings, where an intervention is considered practically indistinguishable from a benchmark unless it changes outcomes beyond a tolerance band. In experimental evaluations of policy changes, for example, policy-makers may be interested in testing the hypothesis that a treatment has an impact on outcomes for any segment of the population beyond a 'distinguishable' or 'economically meaningful' margin.

Two-sided bounded hypotheses are also natural in distributional comparisons. In the two-sample framework, requiring that quantile differences remain inside a band corresponds to requiring that two distributions are close in a strong (sup-norm) sense. This yields finite-sample valid tests of "near distributional equivalence" that are stricter than moment-based notions of closeness.

As in Section 5, we allow the bounds to vary across units as *known* functions of observable characteristics and (at most) one potential outcome. Let $\tau_i = Y_i(1) - Y_i(0)$ denote the individual treatment effect.

Definition 8 (Two-sided bounded treatment effect null)**.** Fix known measurable functions $l : \mathbb{R} \times \mathcal{Z} \to \mathbb{R}$ and $u : \mathbb{R} \times \mathcal{Z} \to \mathbb{R}$ such that $l(y, z) \leq u(y, z)$ for all (y, z). The *two-sided bounded treatment effect null* is

$$H_0^{\ell,u} : \; P\Big(l\big(Y_i(0), Z_i\big) \leq \tau_i \leq u\big(Y_i(0), Z_i\big) \text{ for all } i\Big) = 1.$$

Equivalently, under $H_0^{\ell,u}$ we have the pointwise restrictions

$$Y_i(0) + l\big(Y_i(0), Z_i\big) \;\leq\; Y_i(1) \;\leq\; Y_i(0) + u\big(Y_i(0), Z_i\big) \qquad \text{for all } i$$

almost surely under P.

Constant bounds yield familiar special cases. Taking $l(y, z) \equiv -\delta$ and $u(y, z) \equiv \delta$ for $\delta \geq 0$ gives the "no large effects" null $|\tau_i| \leq \delta$ for all i. More generally, l and u can encode asymmetric tolerances (e.g. harms are tolerated up to 0.1 units while benefits are tolerated up to 0.5 units) or tolerance bands that scale with baseline outcomes (e.g. proportional equivalence).

A two-sided bounded treatment effect null is a conjunction of a lower-bounded and an upper-bounded null. A natural approach is therefore to test both sides, but doing so naively (e.g. via

Bonferroni) can be conservative. Here we instead exploit the same core mechanism as in the one-sided case: construct *extremal completions* of the missing potential outcomes that sit on the *boundary* of the null and dominate the realized data in the direction that makes an effect-increasing statistic as extreme as possible.

Fix realized data $x = (y, d, z)$. For each counterfactual assignment $\tilde{d}$ we construct two imputed outcome vectors:

- an *upper-bound imputation* $\tilde{y}^u$ that corresponds to setting each unit's treatment effect to the *largest* value allowed under the null (the u boundary), and
- a *lower-bound imputation* $\tilde{y}^\ell$ that corresponds to setting each unit's treatment effect to the *smallest* value allowed under the null (the l boundary).

For an effect-increasing statistic T, these two boundary schedules bound the statistic for each assignment: the u-schedule makes T as large as possible, while the l-schedule makes T as small as possible. Consequently, the randomization distribution of

$$\max\Big\{\, T(D, \tilde{y}^u, Z),\ -T(D, \tilde{y}^\ell, Z) \Big\}$$

stochastically dominates the randomization distribution of $|T(D, Y, Z)|$ under any P satisfying $H_0^{\ell,u}$. This yields a computable upper bound on the conditional tail probability required by Theorem 1.

Theorem 3 (Valid Test for Two-Sided Bounded Treatment Effects)**.** *Let $H_0^{\ell,u} : P \in \mathcal{P}_0^{\ell,u}$ be the two-sided bounded treatment effect null in Definition 8. Suppose the conditional assignment distribution $P(D = d \mid \mathcal{S})$ is known and independent of potential outcomes $(Y_i(1), Y_i(0))_i^n$, where $\mathcal{S} \in \sigma(X)$ is a sub-σ-algebra of the data σ-algebra. Let T be an effect-increasing statistic (Definition 7).*

For each realized $x = (y, d, z)$, each $\tilde{d} \in \mathrm{Supp}\{D\}$, and each i, define the upper *and* lower *imputations*

$$\tilde{y}_i^u = \begin{cases} y_i + u(y_i, z_i) & \text{if } d_i = 0,\ \tilde{d}_i = 1, \\ \inf\{\, \eta \in \mathbb{R} :\ \eta + u(\eta, z_i) \geq y_i \,\} & \text{if } d_i = 1,\ \tilde{d}_i = 0, \\ y_i & \text{if } \tilde{d}_i = d_i, \end{cases}$$

$$\tilde{y}_i^\ell = \begin{cases} y_i + l(y_i, z_i) & \text{if } d_i = 0,\ \tilde{d}_i = 1, \\ \sup\{\, \eta \in \mathbb{R} :\ \eta + l(\eta, z_i) \leq y_i \,\} & \text{if } d_i = 1,\ \tilde{d}_i = 0, \\ y_i & \text{if } \tilde{d}_i = d_i. \end{cases}$$

Define the upper-bounding conditional tail function

$$U_{\ell,u}(x, r) := \sum_{\tilde{d} \in \mathrm{Supp}\{D\}} \mathbb{1}\Big\{ T(\tilde{d}, \tilde{y}^u, z) \geq r \quad \text{or} \quad -T(\tilde{d}, \tilde{y}^\ell, z) \geq r \Big\} \cdot P(D = \tilde{d} \mid \mathcal{S}).$$

Then

$$\phi_{\ell,u}(x) \;:=\; \mathbb{1}\Big\{\, U_{\ell,u}\big(x,\ |T(x)|\big) \le \alpha \,\Big\}$$

is a valid level-α test of $H_0^{\ell,u}$.

The proof follows the same logic as Theorem 2: for each counterfactual assignment $\tilde{d}$, the effect-increasing property implies

$$T(\tilde{d}, \tilde{y}^u, z) \;\ge\; T(\tilde{d}, Y(\tilde{d}), z) \;\ge\; T(\tilde{d}, \tilde{y}^\ell, z)$$

under any $P \in \mathcal{P}_0^{\ell,u}$, which yields

$$|T(\tilde{d}, Y(\tilde{d}), z)| \;\le\; \max\big\{T(\tilde{d}, \tilde{y}^u, z),\ -T(\tilde{d}, \tilde{y}^\ell, z)\big\}.$$

Taking conditional probabilities over the known assignment mechanism then produces the upper bound $U_{\ell,u}$ required by Theorem 1.

6.1 Application to Two-Sample Problem

Under the latent-quantile embedding from Section 3, the unit-level treatment effect is the quantile difference at a latent q_i:

$$\tau_i \;=\; F^{-1}(q_i) - G^{-1}(q_i).$$

Thus, a two-sided bounded treatment effect null translates into a uniform two-sided bound on the quantile treatment effect function:

$$l\big(G^{-1}(q)\big) \;\le\; F^{-1}(q) - G^{-1}(q) \;\le\; u\big(G^{-1}(q)\big) \qquad \forall q \in [0,1].$$

This yields the following corollary.

Corollary 2 (Valid Tests for Two-Sided Bounded Quantile Treatment Effects in Two-Sample Problems)**.** *Consider the two-sample setting with $X_i \overset{iid}{\sim} F$ for $i = 1, \ldots, n_X$ and $Y_j \overset{iid}{\sim} G$ for $j = 1, \ldots, n_Y$. Let T be an effect-increasing test statistic. The test $\phi_{\ell,u}$ from Theorem 3 with $l(y,z) = l(y)$ and $u(y,z) = u(y)$ is a valid level-α test of*

$$H_0^{\ell,u}: \quad l\big(G^{-1}(q)\big) \;\le\; F^{-1}(q) - G^{-1}(q) \;\le\; u\big(G^{-1}(q)\big) \qquad \forall q \in [0,1].$$

Proof. The result follows immediately from Theorem 3 under the latent quantile construction used to embed the two-sample problem into the potential outcomes framework. □

As in the one-sided case, inverting this family of level-α tests over δ yields finite-sample valid confidence sets for the worst-case quantile discrepancy $\sup_q |F^{-1}(q) - G^{-1}(q)|$ (equivalently, the Wasserstein-∞ distance under the one-dimensional quantile representation).

6.2 Example 3: Test of Relevant Differences

We revisit the time-preference elicitation study of Brañas-Garza et al. (2023), who investigate whether distributions of elicited preferences are similar across different incentive designs: real payments, probabilistic payment schemes, and hypothetical payments. Rather than testing exact equality of distributions, a sharp null that may be overly stringent, the economically relevant question is whether the distributions are *close enough* that findings under one incentive scheme can be treated as informative about behavior under another. This "no relevant differences" hypothesis can be formalized as a two-sided bound on quantile differences: the distributions are considered practically equivalent if quantile differences remain within a tolerance band $[-\delta, \delta]$ for some research-relevant threshold $\delta > 0$.

In our notation from Section 6.1, let F denote the distribution of elicited preferences under one incentive scheme (e.g., real payments) and G the distribution under another (e.g., hypothetical payments). The null hypothesis of "no relevant differences" is

$$H_0^\delta : \quad \sup_{q \in [0,1]} \left| F^{-1}(q) - G^{-1}(q) \right| \leq \delta,$$

which corresponds to the two-sided bounded treatment effect null from Definition 8 with constant bounds $l(y, z) \equiv -\delta$ and $u(y, z) \equiv \delta$. Under the latent quantile embedding, this is equivalent to testing whether all individual-level treatment effects satisfy $|\tau_i| \leq \delta$ for all i.

Theorem 3 provides a finite-sample valid test of H_0^δ by computing the upper-bounding conditional tail function $U_{\ell,u}(x, |T(x)|)$ over permutations of the pooled sample. For each permuted assignment $\tilde{d}$, we compute the test statistic under both boundary imputations: the upper boundary where all treated outcomes are shifted up by δ (and controls shifted down), and the lower boundary where the shifts are reversed. The p-value is the proportion of permutations for which either boundary statistic exceeds the observed absolute value $|T(x)|$.

The σ-algebra $\mathcal{S}$ is generated by the event that a particular set of quantiles is observed, or in the potential outcome framing that individuals with the same latent quantile q_i are observed. Given this conditioning, the remaining randomness arises solely from the random realization of whether the quantile is being observed from distribution F or G, or equivalently the random assignment of units to treatment and control groups.

We apply our two-sided bounded treatment effect test using the authors' stated equivalence bounds from their own analysis: $\delta = 0.028$ for present bias (β) and $\delta = 0.013$ for long-run patience (δ), corresponding to 0.16 and 0.14 standard deviations respectively. Table 4 reports the results of finite-sample permutation tests (10,000 replications) for the null hypothesis $H_0^\delta : \sup_q |F^{-1}(q) - G^{-1}(q)| \leq \delta$ across three pairwise comparisons of incentive schemes (real vs. hypothetical, real vs. BRIS, and BRIS vs. hypothetical) and two outcomes (present bias β and long-run patience δ). The test statistic is the Kolmogorov-Smirnov supremum of empirical CDF differences, which is effect-increasing.

Our distributional test partially confirms the authors' main conclusion: We find no evidence

Table 4: Bounded Treatment Effect Tests: Spanish Lab Experiment (n=116)

Outcome	Comparison	n_1	n_2	δ	Mean Diff	KS	p-value
Panel A: Present Bias (β)							
	Real vs Hypothetical	40	39	0.028	0.003	0.091	0.946
	Real vs BRIS	40	37	0.028	0.037	0.324	0.019**
	BRIS vs Hypothetical	37	39	0.028	-0.034	0.291	0.047**
Panel B: Long-Run Patience (δ)							
	Real vs Hypothetical	40	39	0.013	0.000	0.092	0.824
	Real vs BRIS	40	37	0.013	0.002	0.087	0.887
	BRIS vs Hypothetical	37	39	0.013	-0.002	0.179	0.277

Notes: Finite-sample permutation tests (10,000 replications) for H_0: $\sup_q |F^{-1}(q) - G^{-1}(q)| \leq \delta$. Equivalence bounds are from Brañas-Garza et al. (2023, p.18): $\delta = 0.028$ for β (0.16 SD) and $\delta = 0.013$ for δ (0.14 SD). Test statistic is Kolmogorov-Smirnov supremum of empirical CDF differences. Rejecting H_0 indicates distributions differ by more than δ. ** p<0.05.

that Real and Hypothetical payment schemes elicit meaningfully different distributions for both parameters (all p > 0.82). However, we find evidence that BRIS differs distributionally from both Real (p = 0.019) and Hypothetical (p = 0.047) for present bias, despite similar means. The Kolmogorov-Smirnov statistics of 0.32 and 0.29 indicate substantial distributional shifts at some quantiles, the empirical CDFs differ by over 30 percentage points. We should conclude that BRIS does not yield equivalent elicited preferences compared to either real or hypothetical payments, at least for present bias. This result shows the potential usefulness of the testing framework for two-sided bounded quantile differences in practical applications.

7 Conclusion

Randomization tests are attractive in empirical work because they deliver exact finite-sample control of Type I error under known assignment mechanisms, but their classical form is largely confined to sharp null hypotheses. This paper shows that a substantial class of economically relevant *non-sharp* hypotheses, namely nulls that impose bounds on individual treatment effects, can nonetheless be tested with finite-sample validity using randomization-style logic. I have shown how finite-sample valid tests can be constructed for one-sided and two-sided bounded treatment effect nulls and can be applied to a variety of applied problems, including testing the monotonicity assumption in instrumental variables designs, constructing confidence bounds on worst-case individual treatment effects, and testing for relevant differences in distributional comparisons.

While this paper establishes finite-sample validity of these tests, it does not treat the question of *power*. Understanding power is an important next step, especially because the tests developed here are deliberately conservative by construction, being calibrated to least favorable boundary configurations. Future work should characterize the potential power of these test more clearly, examining the conditions on test statistics and alternative hypotheses for consistentcy of these tests. Developing procedures to identify power-optimal test statistics for a given hypothesis could

prove useful for application of these tests, and guide researchers in the trade-offs involved with choosing finite-sample validity. One challenge for this work is that finite-sample notions of power optimality for randomization-style procedures have not been widely developed in the literature.

More broadly, Theorem 1, though simple in its insight, has the potential to serve as a foundation for constructing finite-sample tests for a wide array of non-sharp null hypotheses. A promising direction for future research is to extend the boundary-completion and tail-bound logic to hypotheses that place restrictions on specific quantiles of treatment effect distributions, as well as to hypotheses about average treatment effects. Because these are among the most commonly tested and assumed restrictions in empirical econometrics, developing finite-sample valid randomization tests for such targets could bring the benefits of randomization inference to a much larger set of applications in economics, particularly in settings where effective sample sizes are small and asymptotic approximations are fragile.

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A Proofs

A.1 Proof of Lemma 1

Before we get to proving the main lemma, we will actually need to first show that characteristic 1 of conditional distribution holds not just for a fixed Borel set, but also for random Borel sets that are measurable with respect to the conditioning sigma-algebra. I believe similar results are standard in measure theory, but I include it here for completeness.

Lemma 2 (Random Threshold Evaluation)**.** *Let Y be a real-valued random variable on $(\Omega, \mathcal{F}, P)$, let $\mathcal{S} \subseteq \mathcal{F}$ be a sub-σ-algebra, and let $\tilde{\kappa}(\omega, \cdot)$ be a regular conditional distribution of Y given $\mathcal{S}$ under P. If R is $\mathcal{S}$-measurable, then*

$$P(Y \geq R \mid \mathcal{S})(\omega) = \tilde{\kappa}\big(\omega, [R(\omega), \infty)\big) \quad a.s.$$

and similarly

$$P(Y > R \mid \mathcal{S})(\omega) = \tilde{\kappa}\big(\omega, (R(\omega), \infty)\big) \quad a.s.$$

Proof. First suppose R is simple: $R = \sum_{j=1}^{m} r_j \mathbf{1}_{A_j}$ with $A_j \in \mathcal{S}$. Then

$$\mathbf{1}\{Y \geq R\} = \sum_{j=1}^{m} \mathbf{1}_{A_j} \mathbf{1}\{Y \geq r_j\},$$

so by linearity and the pull-out property of conditional expectation,

$$P(Y \geq R \mid \mathcal{S}) = \sum_{j=1}^{m} \mathbf{1}_{A_j} P(Y \geq r_j \mid \mathcal{S}).$$

By the defining property of a regular conditional distribution, $P(Y \geq r_j \mid \mathcal{S})(\omega) = \tilde{\kappa}(\omega, [r_j, \infty))$ a.s., hence

$$P(Y \geq R \mid \mathcal{S})(\omega) = \sum_{j=1}^{m} \mathbf{1}_{A_j}(\omega)\, \tilde{\kappa}(\omega, [r_j, \infty)) = \tilde{\kappa}\big(\omega, [R(\omega), \infty)\big) \quad \text{a.s.}$$

For general $\mathcal{S}$-measurable R, choose simple $R_n \downarrow R$ pointwise. Then $\mathbf{1}\{Y \geq R_n\} \uparrow \mathbf{1}\{Y \geq R\}$, so $P(Y \geq R_n \mid \mathcal{S}) \uparrow P(Y \geq R \mid \mathcal{S})$ a.s. by monotone convergence, and also $[R_n(\omega), \infty) \downarrow [R(\omega), \infty)$ implies $\tilde{\kappa}(\omega, [R_n(\omega), \infty)) \uparrow \tilde{\kappa}(\omega, [R(\omega), \infty))$ by continuity from below. Passing to the limit yields the stated identity for $\geq$.

For $>$, note $\mathbf{1}\{Y > R\} = \lim_{n\to\infty} \mathbf{1}\{Y \geq R + 1/n\}$, hence

$$P(Y > R \mid \mathcal{S}) = \lim_{n\to\infty} P(Y \geq R + 1/n \mid \mathcal{S}) = \lim_{n\to\infty} \tilde{\kappa}(\cdot, [R + 1/n, \infty)) = \tilde{\kappa}(\cdot, (R, \infty)),$$

using continuity from below of measures and $\bigcup_n [R + 1/n, \infty) = (R, \infty)$. □

Lemma 1 (Super-Uniformity of Conditional p-Value, Restated). *Let $T : \mathcal{X} \to \mathbb{R}$ be a test statistic, $\mathcal{S} \subseteq \mathcal{F}^X$ a sub-sigma algebra, and P a probability measure on $(\Omega, \mathcal{F})$. Let $p_{P|\mathcal{S}}(X)$ denote the conditional p-value as defined in Definition 4. Then, for every $\alpha \in [0, 1]$,*

$$P\big(p_{P|\mathcal{S}}(X) \leq \alpha \mid \mathcal{S}\big) \leq \alpha.$$

Proof. Let $Y := T(X)$, and let $\tilde{\kappa}_{P|\mathcal{S}}(\omega, \cdot)$ be a regular conditional distribution of Y given $\mathcal{S}$. Write

$$p_{P|\mathcal{S}}(\omega) := \tilde{\kappa}_{P|\mathcal{S}}\big(\omega, [Y(\omega), \infty)\big).$$

Fix $\alpha \in [0, 1]$. For each ω, define the tail function

$$\bar{F}_\omega(t) := \tilde{\kappa}_{P|\mathcal{S}}\big(\omega, [t, \infty)\big).$$

Define the upper set and its infimum

$$A_\alpha(\omega) := \{t \in \mathbb{R} : \bar{F}_\omega(t) \le \alpha\}, \qquad c_\alpha(\omega) := \inf A_\alpha(\omega) \quad (\inf \emptyset := +\infty).$$

Because $\bar{F}_\omega(\cdot)$ is nonincreasing, $A_\alpha(\omega)$ is either $[c_\alpha(\omega), \infty)$ or $(c_\alpha(\omega), \infty)$, and

$$\{p_{P|\mathcal{S}} \le \alpha\} = \{Y \in A_\alpha(\cdot)\},$$

since $p_{P|\mathcal{S}}(\omega) = \bar{F}_\omega(Y(\omega))$.

We claim c_α is $\mathcal{S}$-measurable. Indeed, for any $a \in \mathbb{R}$,

$$\{c_\alpha < a\} = \bigcup_{q \in \mathbb{Q},\, q<a} \{\bar{F}_\omega(q) \le \alpha\},$$

and $\omega \mapsto \bar{F}_\omega(q) = \tilde{\kappa}_{P|\mathcal{S}}(\omega, [q, \infty))$ is $\mathcal{S}$-measurable for each rational q by the defining property of the r.c.d.

Let

$$I(\omega) := \mathbf{1}\{\bar{F}_\omega(c_\alpha(\omega)) \le \alpha\}.$$

Pointwise in ω,

$$\mathbf{1}\{Y \in A_\alpha(\omega)\} = I(\omega)\, \mathbf{1}\{Y \ge c_\alpha(\omega)\} + (1 - I(\omega))\, \mathbf{1}\{Y > c_\alpha(\omega)\}.$$

Taking conditional expectations given $\mathcal{S}$ gives

$$P(p_{P|\mathcal{S}} \le \alpha \mid \mathcal{S}) = I \cdot P(Y \ge c_\alpha \mid \mathcal{S}) + (1 - I) \cdot P(Y > c_\alpha \mid \mathcal{S}).$$

Apply Lemma 2 with $R = c_\alpha$ (which is $\mathcal{S}$-measurable). Then, a.s.,

$$P(Y \ge c_\alpha \mid \mathcal{S})(\omega) = \tilde{\kappa}_{P|\mathcal{S}}\big(\omega, [c_\alpha(\omega), \infty)\big) = \bar{F}_\omega(c_\alpha(\omega)),$$

and

$$P(Y > c_\alpha \mid \mathcal{S})(\omega) = \tilde{\kappa}_{P|\mathcal{S}}\big(\omega, (c_\alpha(\omega), \infty)\big).$$

Hence, a.s.,

$$P(p_{P|\mathcal{S}} \le \alpha \mid \mathcal{S})(\omega) = I(\omega)\, \bar{F}_\omega(c_\alpha(\omega)) + (1 - I(\omega))\, \tilde{\kappa}_{P|\mathcal{S}}\big(\omega, (c_\alpha(\omega), \infty)\big).$$

On $\{I = 1\}$, $\bar{F}_\omega(c_\alpha(\omega)) \le \alpha$ by definition of I. On $\{I = 0\}$, we have $c_\alpha(\omega) \notin A_\alpha(\omega)$ but $c_\alpha(\omega) = \inf A_\alpha(\omega)$, so $c_\alpha(\omega) + 1/n \in A_\alpha(\omega)$ for all n, i.e.

$$\tilde{\kappa}_{P|\mathcal{S}}\big(\omega, [c_\alpha(\omega) + 1/n, \infty)\big) \le \alpha \quad \forall n.$$

By continuity from below and $\bigcup_n [c_\alpha + 1/n, \infty) = (c_\alpha, \infty)$,

$$\tilde{\kappa}_{P|\mathcal{S}}\big(\omega, (c_\alpha(\omega), \infty)\big) = \lim_{n\to\infty} \tilde{\kappa}_{P|\mathcal{S}}\big(\omega, [c_\alpha(\omega) + 1/n, \infty)\big) \le \alpha.$$

Therefore $P(p_{P|\mathcal{S}} \le \alpha \mid \mathcal{S}) \le \alpha$ a.s. □

A.2 Proof of Proposition 1

Proposition 1 (Group invariance is sufficient for Theorem 1, Restated). *Suppose Condition 1 holds for a null hypothesis $H_0 : P \in \mathcal{P}_0$ with a* finite *group G of measurable transformations acting on $\mathcal{X}$. Define the (set-valued) orbit map*

$$\mathrm{orb}(x) := \{g(x) : g \in G\} \subseteq \mathcal{X},$$

and let $\mathcal{S} := \sigma(\mathrm{orb}(X))$ denote the σ-algebra generated by the orbit of the realized data.

Then the function

$$U(x, r) \;:=\; \frac{1}{|G|} \sum_{g\in G} \mathbb{1}\{\, T(g(x)) \ge r \,\}, \qquad (x, r) \in \mathcal{X} \times \mathbb{R},$$

is known and measurable, and satisfies, for every $P \in \mathcal{P}_0$ and $r \in \mathbb{R}$,

$$U(x, r) \;=\; \bar{F}_{P|\mathcal{S}}(x, r) \qquad \textit{for } P^X\textit{-a.e. } x.$$

Proof. Fix any $P \in \mathcal{P}_0$ and $r \in \mathbb{R}$, and write $m := |G|$. We interpret Condition 1 in the standard sense that

$$g(X) \overset{d}{=} X \quad \text{under } P \qquad \forall g \in G,$$

equivalently $P(g(X) \in A) = P(X \in A)$ for all $A \in \mathcal{F}^X$.

U **is measurable:** For each fixed $g \in G$, the map $(x, r) \mapsto T(g(x))$ is measurable, hence so is $(x, r) \mapsto \mathbb{1}\{T(g(x)) \ge r\}$. Because G is finite, U is a finite average of measurable maps, hence measurable.

$U(\cdot, r)$ **is $\mathcal{S}$-measurable:** For any $h \in G$ and $x \in \mathcal{X}$,

$$U(h(x), r) = \frac{1}{m} \sum_{g\in G} \mathbb{1}\{T(g(h(x))) \ge r\}.$$

Since $g \mapsto g \circ h$ is a bijection of G, this equals $\frac{1}{m}\sum_{k\in G} \mathbb{1}\{T(k(x)) \ge r\} = U(x, r)$. Thus $U(\cdot, r)$ is constant on orbits.

Identify $U(X, r)$ as the conditional tail probability: Let $Y_r := \mathbb{1}\{T(X) \ge r\}$. We will show that $U(X, r)$ is a version of $E_P[Y_r \mid \mathcal{S}] = P(T(X) \ge r \mid \mathcal{S})$. Let $A \in \mathcal{S}$. Then $A = \{\mathrm{orb}(X) \in B\}$

for some B. Using $\text{orb}(g(x)) = \text{orb}(x)$ for all $g \in G$ and $x \in \mathcal{X}$, we have

$$\mathbb{1}_A = \mathbb{1}\{\text{orb}(X) \in B\} = \mathbb{1}\{\text{orb}(g(X)) \in B\} \qquad P\text{-a.s.}$$

Hence

$$\begin{aligned} E_P[U(X,r)\,\mathbb{1}_A] &= \frac{1}{m}\sum_{g\in G} E_P[\mathbb{1}\{T(g(X)) \geq r\}\,\mathbb{1}_A] \\ &= \frac{1}{m}\sum_{g\in G} E_P[\mathbb{1}\{T(g(X)) \geq r\}\,\mathbb{1}\{\text{orb}(g(X)) \in B\}] \\ &= \frac{1}{m}\sum_{g\in G} E_P[f(g(X))]\,, \end{aligned}$$

where $f : \mathcal{X} \to \mathbb{R}$ is $f(x) := \mathbb{1}\{T(x) \geq r\}\,\mathbb{1}\{\text{orb}(x) \in B\}$. By group invariance, $g(X) \overset{d}{=} X$ under P, so $E_P[f(g(X))] = E_P[f(X)]$ for each $g \in G$. Therefore,

$$E_P[U(X,r)\,\mathbb{1}_A] = E_P[f(X)] = E_P[\mathbb{1}\{T(X) \geq r\}\,\mathbb{1}\{\text{orb}(X) \in B\}] = E_P[Y_r\,\mathbb{1}_A].$$

Since $U(X,r)$ is $\mathcal{S}$-measurable and satisfies $E_P[U(X,r)\mathbb{1}_A] = E_P[Y_r\mathbb{1}_A]$ for all $A \in \mathcal{S}$, it follows that

$$U(X,r) = E_P[Y_r \mid \mathcal{S}] = P(T(X) \geq r \mid \mathcal{S}) = \bar{F}_{P|\mathcal{S}}(X,r) \qquad P\text{-a.s.}$$

□

A.3 Proof of Theorem 2

Theorem 2 (Valid Test for One-Sided Bounded Treatment Effects, Restated). *Let $H_0^\ell : P \in \mathcal{P}_0^\ell$ be the one-sided lower bounded treatment effect null as in Definition 5, or let $H_0^u : P \in \mathcal{P}_0^u$ be the one-sided upper bounded treatment effect null as in Definition 6. Suppose the conditional distribution $P(D = d \mid \mathcal{S})$ is known, where $\mathcal{S}$ is a sub-σ-algebra of the data σ-algebra. Let T be an* effect-increasing *test statistic (Definition 7).*

(i) Upper bounded null H_0^u: *Define*

$$U_u(x,r) = \sum_{\tilde{d}\in\text{Supp}\{D\}} \mathbb{1}\Big\{T\big(\tilde{d}, \tilde{y}^u, z\big) \geq r\Big\} \cdot P\big(D = \tilde{d} \mid \mathcal{S}\big),$$

$$\tilde{y}_i^u = \begin{cases} y_i + u(y_i, z_i) & \text{if } d_i = 0, \tilde{d}_i = 1, \\ \inf\{y : y + u(y, z_i) \geq y_i\} & \text{if } d_i = 1, \tilde{d}_i = 0, \\ y_i & \text{if } \tilde{d}_i = d_i. \end{cases}$$

Then $\phi_u(x) = \mathbb{1}\{U_u(x, T(x)) \leq \alpha\}$ is a valid level-α test of H_0^u.

(ii) Lower bounded null H_0^ℓ: *Define the* effect-decreasing *statistic $T^-(d,y,z) := -T(d,y,z)$,*

and

$$U_\ell(x,r) = \sum_{\tilde{d}\in \mathrm{Supp}\{D\}} \mathbb{1}\Big\{T^-\big(\tilde{d},\tilde{y}^\ell,z\big)\geq r\Big\}\cdot P\big(D=\tilde{d}\mid \mathcal{S}\big),$$

$$\tilde{y}_i^\ell = \begin{cases} y_i + l(y_i,z_i) & \text{if } d_i=0,\tilde{d}_i=1,\\ \sup\{y: y+l(y,z_i)\geq y_i\} & \text{if } d_i=1,\tilde{d}_i=0,\\ y_i & \text{if } \tilde{d}_i=d_i. \end{cases}$$

Then $\phi_\ell(x)=\mathbb{1}\{U_\ell(x,T^-(x))\leq\alpha\}$ is a valid level-α test of H_0^ℓ.

Proof. We verify the sufficient condition in Theorem 1.

(i) Upper-bounded null H_0^u. Fix any $P\in\mathcal{P}_0^u$ and $r\in\mathbb{R}$. Since $P\in\mathcal{P}_0^u$,

$$Y_i(1)\leq Y_i(0)+u\big(Y_i(0),Z_i\big)\qquad \text{for all } i\quad P\text{-a.s.}$$

For each realized $x=(d,y,z)$, define the *extremal* (least-favorable) completion $(Y^u(0),Y^u(1))$ by

$$Y_i^u(0)=\begin{cases} y_i, & d_i=0,\\ \inf\{\,\eta\in\mathbb{R}:\ \eta+u(\eta,z_i)\geq y_i\,\}, & d_i=1,\end{cases}\qquad Y_i^u(1)=\begin{cases} y_i+u(y_i,z_i), & d_i=0,\\ y_i, & d_i=1.\end{cases}$$

For any counterfactual assignment $\tilde{d}$, let

$$\tilde{y}^u(\tilde{d}) := \tilde{d}\circ Y^u(1)+(\mathbb{1}-\tilde{d})\circ Y^u(0),$$

which agrees with the piecewise definition of $\tilde{y}_i^u$ in the theorem statement.

Claim (coordinatewise dominance). Under H_0^u, for each i,

$$Y_i^u(1)\geq Y_i(1)\quad\text{and}\quad Y_i^u(0)\leq Y_i(0)\qquad P\text{-a.s.}$$

Indeed, if $d_i=0$ then $y_i=Y_i(0)$ and $Y_i^u(1)=Y_i(0)+u(Y_i(0),z_i)\geq Y_i(1)$ by the null. If $d_i=1$ then $y_i=Y_i(1)$ and $Y_i(0)$ belongs to the set $\{\eta:\eta+u(\eta,z_i)\geq y_i\}$ (again by the null), hence $Y_i^u(0)\leq Y_i(0)$ by definition of the infimum.

Now fix any $\tilde{d}\in\mathrm{Supp}\{D\}$. By the claim, relative to the true schedule, $\tilde{y}^u(\tilde{d})$ weakly increases treated components ($\tilde{d}_i=1$) and weakly decreases control components ($\tilde{d}_i=0$). Since T is effect-increasing (Definition 7),

$$T\big(\tilde{d},\tilde{y}^u(\tilde{d}),z\big)\ \geq\ T\big(\tilde{d},Y(\tilde{d}),z\big)\qquad P\text{-a.s.}$$

Hence for every $\tilde{d}$ and r,

$$\mathbb{1}\Big\{T\big(\tilde{d}, Y(\tilde{d}), z\big) \geq r\Big\} \;\leq\; \mathbb{1}\Big\{T\big(\tilde{d}, \tilde{y}^u(\tilde{d}), z\big) \geq r\Big\} \qquad P\text{-a.s.}$$

Conditioning on $\mathcal{S}$, and using that $P(D = \tilde{d} \mid \mathcal{S})$ is known by assumption, we obtain

$$\begin{aligned}
\bar{F}_{P|\mathcal{S}}(X, r) &:= P(T(X) \geq r \mid \mathcal{S}) \\
&= \sum_{\tilde{d} \in \mathrm{Supp}\{D\}} \mathbb{1}\Big\{T\big(\tilde{d}, Y(\tilde{d}), Z\big) \geq r\Big\} P(D = \tilde{d} \mid \mathcal{S}) \\
&\leq \sum_{\tilde{d} \in \mathrm{Supp}\{D\}} \mathbb{1}\Big\{T\big(\tilde{d}, \tilde{y}^u(\tilde{d}), Z\big) \geq r\Big\} P(D = \tilde{d} \mid \mathcal{S}) \;=\; U_u(X, r),
\end{aligned}$$

where the final equality is exactly the definition of U_u. Thus $U_u(\cdot, r) \geq \bar{F}_{P|\mathcal{S}}(\cdot, r)$ P-a.s. for every r. By Theorem 1, $\phi_u(x) = \mathbb{1}\{U_u(x, T(x)) \leq \alpha\}$ is level-α for H_0^u.

(ii) Lower-bounded null H_0^ℓ. Fix any $P \in \mathcal{P}_0^\ell$ and define $T^-(d, y, z) := -T(d, y, z)$. Under $P \in \mathcal{P}_0^\ell$,

$$Y_i(1) \geq Y_i(0) + \ell\big(Y_i(0), Z_i\big) \qquad \text{for all } i \quad P\text{-a.s.}$$

For each realized $x = (d, y, z)$, define the extremal completion $(Y^\ell(0), Y^\ell(1))$ by

$$Y_i^\ell(0) = \begin{cases} y_i, & d_i = 0, \\ \sup\{\, \eta \in \mathbb{R} :\; \eta + \ell(\eta, z_i) \leq y_i \,\}, & d_i = 1, \end{cases} \qquad Y_i^\ell(1) = \begin{cases} y_i + \ell(y_i, z_i), & d_i = 0, \\ y_i, & d_i = 1. \end{cases}$$

For any $\tilde{d}$, let $\tilde{y}^\ell(\tilde{d}) := \tilde{d} \circ Y^\ell(1) + (\mathbb{1} - \tilde{d}) \circ Y^\ell(0)$.

Claim (reverse dominance). Under H_0^ℓ, for each i,

$$Y_i^\ell(1) \leq Y_i(1) \quad \text{and} \quad Y_i^\ell(0) \geq Y_i(0) \qquad P\text{-a.s.}$$

If $d_i = 0$, then $y_i = Y_i(0)$ and $Y_i^\ell(1) = Y_i(0) + \ell(Y_i(0), z_i) \leq Y_i(1)$ by the null. If $d_i = 1$, then $y_i = Y_i(1)$ and the true $Y_i(0)$ satisfies $Y_i(0) + \ell(Y_i(0), z_i) \leq y_i$, so $Y_i(0)$ lies in the set defining the supremum, hence $Y_i^\ell(0) \geq Y_i(0)$.

Fix any $\tilde{d}$. Relative to the true schedule, $\tilde{y}^\ell(\tilde{d})$ weakly decreases treated components and weakly increases control components, so effect-increasing T implies

$$T\big(\tilde{d}, \tilde{y}^\ell(\tilde{d}), z\big) \leq T\big(\tilde{d}, Y(\tilde{d}), z\big), \quad \text{equivalently} \quad T^-\big(\tilde{d}, \tilde{y}^\ell(\tilde{d}), z\big) \geq T^-\big(\tilde{d}, Y(\tilde{d}), z\big) \qquad P\text{-a.s.}$$

Therefore, as in part (i), for every r,

$$P\big(T^-(X) \geq r \mid \mathcal{S}\big) \leq \sum_{\tilde{d} \in \mathrm{Supp}\{D\}} \mathbb{1}\Big\{T^-\big(\tilde{d}, \tilde{y}^\ell(\tilde{d}), Z\big) \geq r\Big\} P(D = \tilde{d} \mid \mathcal{S}) \;=\; U_\ell(X, r).$$

So $U_\ell(\cdot, r) \geq \bar{F}^-_{P|\mathcal{S}}(\cdot, r)$ P-a.s. for every r, where $\bar{F}^-$ is the conditional tail probability for T^-. By

Theorem 1 applied to T^- and U_ℓ, the test $\phi_\ell(x) = \mathbb{1}\{U_\ell(x, T^-(x)) \le \alpha\}$ is level-α for H_0^ℓ. □

A.4 Proof of Theorem 3

Theorem 3 (Valid Test for Two-Sided Bounded Treatment Effects, Restated). *Let $H_0^{\ell,u} : P \in \mathcal{P}_0^{\ell,u}$ be the two-sided bounded treatment effect null in Definition 8. Suppose the conditional assignment distribution $P(D = d \mid \mathcal{S})$ is known, where $\mathcal{S}$ is a sub-σ-algebra of the data σ-algebra. Let T be an effect-increasing statistic (Definition 7).*

For each realized $x = (y, d, z)$, each $\tilde{d} \in \text{Supp}\{D\}$, and each i, define the upper *and* lower *imputations*

$$\tilde{y}_i^u = \begin{cases} y_i + u(y_i, z_i) & \text{if } d_i = 0,\ \tilde{d}_i = 1, \\ \inf\{\, \eta \in \mathbb{R} :\ \eta + u(\eta, z_i) \ge y_i \,\} & \text{if } d_i = 1,\ \tilde{d}_i = 0, \\ y_i & \text{if } \tilde{d}_i = d_i, \end{cases}$$

$$\tilde{y}_i^\ell = \begin{cases} y_i + l(y_i, z_i) & \text{if } d_i = 0,\ \tilde{d}_i = 1, \\ \sup\{\, \eta \in \mathbb{R} :\ \eta + l(\eta, z_i) \le y_i \,\} & \text{if } d_i = 1,\ \tilde{d}_i = 0, \\ y_i & \text{if } \tilde{d}_i = d_i. \end{cases}$$

Define the upper-bounding conditional tail function

$$U_{\ell,u}(x, r) := \sum_{\tilde{d} \in \text{Supp}\{D\}} \mathbb{1}\Big\{ T(\tilde{d}, \tilde{y}^u, z) \ge r \quad or \quad -T(\tilde{d}, \tilde{y}^\ell, z) \ge r \Big\} \cdot P(D = \tilde{d} \mid \mathcal{S}).$$

Then

$$\phi_{\ell,u}(x) \ :=\ \mathbb{1}\Big\{\, U_{\ell,u}\big(x,\ |T(x)|\big) \le \alpha \,\Big\}$$

is a valid level-α test of $H_0^{\ell,u}$.

Proof. Fix any $P \in \mathcal{P}_0^{l,u}$ and $r \in \mathbb{R}$. Write the observed data as $X = (D, Y, Z)$ with $Y = Y(D)$, and define the two-sided test statistic

$$T_{\text{abs}}(x) \ :=\ |T(x)|.$$

We will verify the hypothesis of Theorem 1 with T_{abs} and $U(x, r) = \pi_{l,u}(x, r)$.

For each realized $x = (d, y, z)$, define two completed potential-outcome schedules $(Y^U(0), Y^U(1))$ and $(Y^L(0), Y^L(1))$ that are consistent with x by

$$Y_i^U(0) = \begin{cases} y_i, & d_i = 0, \\ \inf\{\eta \in \mathbb{R} :\ \eta + u(\eta, z_i) \ge y_i\}, & d_i = 1, \end{cases} \qquad Y_i^U(1) = \begin{cases} y_i + u(y_i, z_i), & d_i = 0, \\ y_i, & d_i = 1, \end{cases}$$

and

$$Y_i^L(0) = \begin{cases} y_i, & d_i = 0, \\ \sup\{\eta \in \mathbb{R} : \ \eta + l(\eta, z_i) \le y_i\}, & d_i = 1, \end{cases} \qquad Y_i^L(1) = \begin{cases} y_i + l(y_i, z_i), & d_i = 0, \\ y_i, & d_i = 1. \end{cases}$$

For any counterfactual assignment $\tilde{d}$, let

$$\tilde{y}^U(\tilde{d}) := \tilde{d} \circ Y^U(1) + (\mathbb{1} - \tilde{d}) \circ Y^U(0), \qquad \tilde{y}^L(\tilde{d}) := \tilde{d} \circ Y^L(1) + (\mathbb{1} - \tilde{d}) \circ Y^L(0),$$

so that $(\tilde{d}, \tilde{y}^U(\tilde{d}), z)$ and $(\tilde{d}, \tilde{y}^L(\tilde{d}), z)$ coincide with the $\tilde{x}^U$ and $\tilde{x}^L$ objects in the theorem statement.

Because $P \in \mathcal{P}_0^{l,u}$, we have for each i,

$$Y_i(0) + l\big(Y_i(0), Z_i\big) \ \le \ Y_i(1) \ \le \ Y_i(0) + u\big(Y_i(0), Z_i\big) \qquad P\text{-a.s.}$$

From the definitions above, it follows (componentwise) that

$$Y^U(1) \ \ge \ Y(1), \quad Y^U(0) \ \le \ Y(0), \qquad \text{and} \qquad Y^L(1) \ \le \ Y(1), \quad Y^L(0) \ \ge \ Y(0) \qquad P\text{-a.s.}$$

(The control case $d_i = 0$ is immediate from the inequalities; the treated case $d_i = 1$ follows because the true $Y_i(0)$ belongs to the feasible sets defining the infimum/supremum.)

Fix any $\tilde{d}$. By Step 2, relative to the true schedule, $\tilde{y}^U(\tilde{d})$ weakly increases treated components and weakly decreases control components, while $\tilde{y}^L(\tilde{d})$ does the reverse. Since T is effect-increasing (Definition 7), we obtain

$$T\big(\tilde{d}, \tilde{y}^U(\tilde{d}), z\big) \ \ge \ T\big(\tilde{d}, Y(\tilde{d}), z\big) \ \ge \ T\big(\tilde{d}, \tilde{y}^L(\tilde{d}), z\big) \qquad P\text{-a.s.}$$

Consequently, for every $r \in \mathbb{R}$,

$$\mathbb{1}\Big\{\big|T\big(\tilde{d}, Y(\tilde{d}), z\big)\big| \ge r\Big\} \le \mathbb{1}\Big\{T\big(\tilde{d}, \tilde{y}^U(\tilde{d}), z\big) \ge r \ \text{ or } \ -T\big(\tilde{d}, \tilde{y}^L(\tilde{d}), z\big) \ge r\Big\}$$

because $\{|a| \ge r\} = \{a \ge r\} \cup \{-a \ge r\}$ and $T(\tilde{d}, Y(\tilde{d}), z)$ is sandwiched between the two extremal values.

Under the known assignment mechanism (conditional on $\mathcal{S}$), the conditional tail probability of $T_{\text{abs}}(X) = |T(X)|$ can be written as

$$\bar{F}^{\text{abs}}_{P|\mathcal{S}}(X, r) := P(|T(X)| \ge r \mid \mathcal{S}) = \sum_{\tilde{d} \in \text{Supp}\{D\}} \mathbb{1}\Big\{\big|T\big(\tilde{d}, Y(\tilde{d}), Z\big)\big| \ge r\Big\} \, P(D = \tilde{d} \mid \mathcal{S}),$$

and applying the indicator inequality from Step 3 term-by-term yields

$$\bar{F}^{\text{abs}}_{P|\mathcal{S}}(X, r) \ \le \sum_{\tilde{d} \in \text{Supp}\{D\}} \mathbb{1}\Big\{T(\tilde{x}^U) \ge r \ \text{ or } \ -T(\tilde{x}^L) \ge r\Big\} \, P(D = \tilde{d} \mid \mathcal{S}) \ = \ \pi_{l,u}(X, r),$$

where $\tilde{x}^U$ and $\tilde{x}^L$ are the upper/lower imputed datasets under assignment $\tilde{d}$.

Thus, for every r, $\pi_{l,u}(X,r)$ is a known measurable function satisfying $\pi_{l,u}(X,r) \geq \bar{F}^{\mathrm{abs}}_{P|\mathcal{S}}(X,r)$ P-a.s. for all $P \in \mathcal{P}_0^{l,u}$. By Theorem 1 applied to the statistic $T_{\mathrm{abs}}(x) = |T(x)|$ and the bound $U(x,r) = \pi_{l,u}(x,r)$, the test

$$\phi_{l,u}(x) \;=\; \mathbb{1}\big\{\pi_{l,u}(x, |T(x)|) \leq \alpha\big\}$$

is a level-α test of $H_0^{l,u}$. □